# LINEAR ALGEBRA

Textbook LA → David C. Lay (4th Ed)

Reference Book → Gilbert Strang (best book) 18.065
Divid Pool
Howard Anton

Mark Distribution:-

| | |
|---|---|
| Q | 10 → 12/13 (every monday) |
| A | 15 |
| S1 | 15 |
| S2 | 15 |
| F | 45 |

⇒ linear algebra & learning from data.

* System of linear equations
* determinant
* vector spaces (abstract)

Ch 5 (imp) Eigenvalues & Eigenvectors

* inner product + orthoganility
* SVD (Single Value Decomposition)
* Tensor Analysis

## Linear Equations

$ax + by = c$ , $a, b, c \in R$

↓ line in $R^2$

$abc \neq 0$ simultaneously

$ax + by + cz = d$

↓ plane in $R^3$

$R^2 = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} : x, y \in R \right\}$

$R^3 = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} : x, y, z \in R^3 \right\}$

$R^n$ : set of all possible n-ordered tuples

# the flat land

→ parametric equation determine the dimensions of shape.

→ the number of variables == dimension of object.

circle, square, rect ⟶ 1D

sphere, cube, ⟶ 2D

any 1D obj can lie in 2D, 3D ....

and any 2D can lie in 3D ....

⇒ Stefan Boyd book.

linear equation for n variable.

$$a_1 x_1 + a_2 x_2 + \cdots + a_n x_n = b$$

vector in CS is a feature vector.

$$a\hat{i} + b\hat{j} + c\hat{k} + d\hat{u} = ?$$

(feature)

Oxford → mathematics by AI

→ complex analysis

→ convex optimization

## Linear System

Two or more linear equation is a linear system

$$ax + by = c_1$$
$$a_1x + b_2y = c_2$$

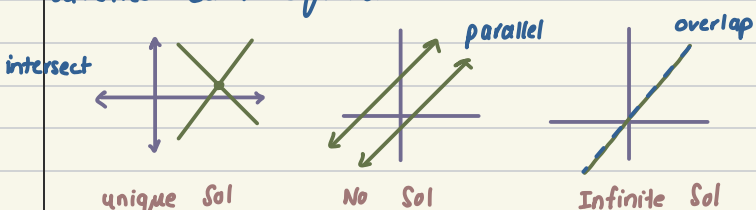
⇒ there is no relation between no of unknowns & no of equation

square ⇒ (equal number) over-determined (eq < var)

rect ⇒ (not equal) under-determined (eq > var)

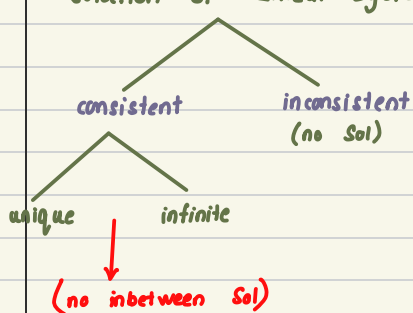
## Solution

satisfies each equation




unique Sol No Sol Infinite Sol

Solution of Linear System




(no inbetween Sol)

Gauss - Elimination → Echelon form

Gauss - Jordon Elimination → Reduced Echelon form

## Gauss Elimination method:-

(i) $x + y = 7$
$2x + 5y = 8$

(ii) $2x + 5y = 8$
$x + y = 7$

(iii) $4x + 10y = 8$
$x + y = 7$

**Row operation**
- → interchange (swap)
- → scaling
- → row replacement

(making 2 equation same & subtracting wor)

$x + 4y + 7z = 7$
$2x + 5y + 9z = 8$
$3x + 5y + 5z = 9$

### Diagonal matrix:-

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 9 \\ 3 & 5 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 7 \\ 8 \\ 9 \end{bmatrix}$$

A X B

### triangular matrix:-

upper triangle ←
$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{bmatrix}$$

lower triangle →
$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 2 & 0 \\ 1 & 4 & 3 \end{bmatrix}$$

### augmented matrix:-

$$A_b = \begin{bmatrix} 1 & 4 & 7 & 7 \\ 2 & 5 & 9 & 8 \\ 3 & 5 & 5 & 9 \end{bmatrix}$$

### equivilent system:-

(having same solution)

$$\begin{bmatrix} 1 & 4 & 7 & 7 \\ 2 & 5 & 9 & 8 \\ 3 & 5 & 5 & 9 \end{bmatrix}$$

$R_2 + (-2)R_1$ , $R_3 - (-3)R_1$

$$\begin{bmatrix} 1 & 4 & 7 & 7 \\ 0 & -3 & -5 & -6 \\ 0 & -7 & -16 & -12 \end{bmatrix}$$

$R_3 + \left(\frac{-7}{3}\right)R_2$

$$\begin{bmatrix} 1 & 4 & 7 & 7 \\ 0 & -3 & -5 & -6 \\ 0 & 0 & -13/3 & 6/3 \end{bmatrix}$$

make equations: -

$-3y - 5z = -6$ , $-\frac{13}{3}z = \frac{6}{3}$

$-3y = -6 + 5\left(\frac{-6}{13}\right)$ ← $z = -6/13$

$y = 2 + \frac{5}{-3}\left(\frac{-6}{13}\right)$ $\quad y = \frac{36}{13}$

①

$\Rightarrow$ $x = u$
$y = v$
$z = 4/3 - 4/3 - 2u$.

$u, v \in \mathbb{R}$

②

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} u \\ v \\ 4/3 - 4/3 - 2/3 u \end{pmatrix}$$

$$= \begin{pmatrix} u \\ 0 \\ -4/3 \end{pmatrix} + \begin{pmatrix} 0 \\ v \\ -2v/3 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 4/3 \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = u \begin{pmatrix} 1 \\ 0 \\ -4/3 \end{pmatrix} + v \begin{pmatrix} 0 \\ 1 \\ -2/3 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 4/3 \end{pmatrix}$$

$x + 4y + 7z = 7$

$x = 7 - 4y - 7z$

$x = 7 - 4 \left(16$

A matrix is said to be an Echelon form.

* if there is an 0, but should be at bottom of the matrix

– leading entry:-

(First non-zero entry)

* The leading entry of any row should be to the right of the preceding row.

$$\begin{matrix} 1 & 4 & 7 & 7 \\ 0 & -3 & -5 & -6 \end{matrix}$$

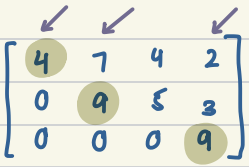
when it matrix is in Echelon form

* leading entry is called pivot position.
* the column containing pivot position is called pivot column.

pivot columns

$$\begin{bmatrix} 4 & 4 & 7 & 9 \\ 0 & 5 & 8 & 8 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

* any linear system is **inconsistent** if and only if last column is the pivot column.



$(0 = 9) \rightarrow$ inconsistent

→ **basic variable** :-
The variable that corresponds to pivot columns

→ **free variable**:-
→ The remaining are free variables.
(not corresponding to pivot cols)

→ having all basic variable ⟶ unique solution
→ having free variable ⟶ infinite solution

1 free variable ⟶ intersecting plane (line)
2 free variable ⟶ intersecting surface (overlaping plane)

$$\begin{bmatrix} 4 & 4 & 7 & 9 \\ 0 & 5 & 8 & 8 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$5y + 8y = 8$

$y = \frac{8}{5} - \frac{8}{5}t$

$4x + 4y + 7z = 9$

$x = \frac{13}{10} - \frac{67}{10}t$

eq of line

$x = x_0 + at$

$y = y_0 + bt$

$z = z_0 + ct$

$x = \frac{13}{10} - \frac{67}{10}t$

$y = \frac{8}{5} - \frac{8}{5}t$

$z = t$

$x = 9 - 3u + 5v$

$y = u$

$z = v$

eq of plane

## Why Echelon forms?

* works on square system
* determinant = 0, which means no Sol / Infinite Sol
* in echelon form, the number of multiplications are $O(n^3)$, because its computational cost is way better than that of cramers / inversion method. (it is n!)
* it gives exact solution, the better ones are approximate

(O notations ⇒ find)

* back substitution $(On^2)$

## Big O, small o, Omega

↳ when data reaches infinite level
→ we check behaviour at infinity

## Reduced Echelon Form:-

* if it is in echelon form
* if all pivot positions are 1.
* pivot is only non zero entity in that column.
* we can directly find Solution.

## Vector Equation:-

↳ ordered list of numbers / features.

vector equation

$$\begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow$$ 2 vectors in $\mathbb{R}^2$

→ 2 data Points having (2 features)

$$x\begin{pmatrix} 1 \\ 2 \end{pmatrix} + y\begin{pmatrix} 3 \\ 4 \end{pmatrix}$$

$$\begin{matrix} v_1 & v_2 & v_3 \\ \begin{pmatrix} 1 & 4 & 0 \\ 2 & 5 & 1 \\ 3 & 6 & 2 \end{pmatrix} & \begin{pmatrix} x \\ y \\ z \end{pmatrix} \begin{matrix} \to a \\ \to b \\ \to c \end{matrix} \end{matrix}$$

↓ ↓ ↓
these are vectors

→ this is weightage assigned to features.

⇒ in real life each row is a feature so, they take transpose

* **Linear Combinations:-**

multiply scalars with vectors.

$v_1, v_2$

$c_1v_1 + c_2v_2 + c_3v_3 \longrightarrow$ linear combination of $v_1, v_2$

Linear

$v_1, v_2 \ldots\ldots v_n$

$c_1v_1 + c_2v_2 + \ldots\ldots c_nv_n$

$v_i$'s $\Rightarrow$ $i = 1 \ldots\ldots n$

* **Affine Combination:-**

$$\sum_{i=1}^{n} c_i = 1$$

* **Convex Combination:-**

$$\sum_{i=1}^{n} c_i = 1$$

$c_i \geq 0$

↳ they are used to find minimas, to find convex functions

* **Span:-**

span of non-zero vector → line
span of zero vector → point
span of multiple vectors → line
span of vectors (not multiple) → plane

span $\left\{\begin{pmatrix}0\\0\end{pmatrix}\right\}$ → point

Span $\left\{\begin{pmatrix}1\\1\end{pmatrix}\right\}$ ↙ $y = x$

Span $\left\{\begin{pmatrix}1\\2\end{pmatrix}\right\}$ ↙ $y = 2x$

Span $\left\{\begin{pmatrix}1\\2\end{pmatrix}, \begin{pmatrix}2\\4\end{pmatrix}\right\}$

Span $\left\{\begin{pmatrix}1\\2\\3\end{pmatrix}, \begin{pmatrix}3\\4\\7\end{pmatrix}\right\}$

Span $= \left\{\begin{pmatrix}1\\2\\3\end{pmatrix}\right\}$ $x = t$, $y = 2t$, $z = 3t$

$Ti + Qj$

$T\begin{pmatrix}1\\0\end{pmatrix}i + Q\begin{pmatrix}0\\1\end{pmatrix}j$

$v_1$ & $v_2$ if they are not multiple ↓ they generate whole of $\mathbb{R}^2$

What is diff between $\mathbb{R}_2$ & $\mathbb{R}_3$

$\mathbb{R}_2$ = set of ordered pairs

only 1 plane

$\mathbb{R}_3$ = set of ordered triple

infinite planes

⇒ find equation in $\mathbb{R}_2$

Zero vector:- (null vector)

→ zero magnitude
→ arbitrary direction. (every direction)
→ this vector is multiple of every vector.

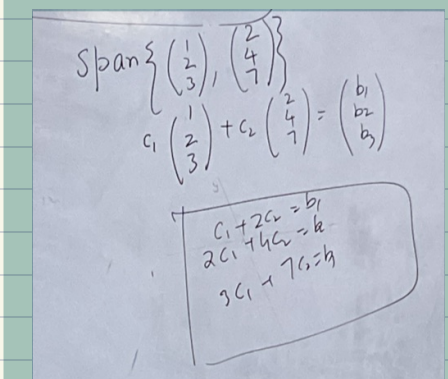
→ $\text{Span}\left\{\begin{pmatrix}1\\2\end{pmatrix}, \begin{pmatrix}3\\4\end{pmatrix}\right\} = \mathbb{R}^2$

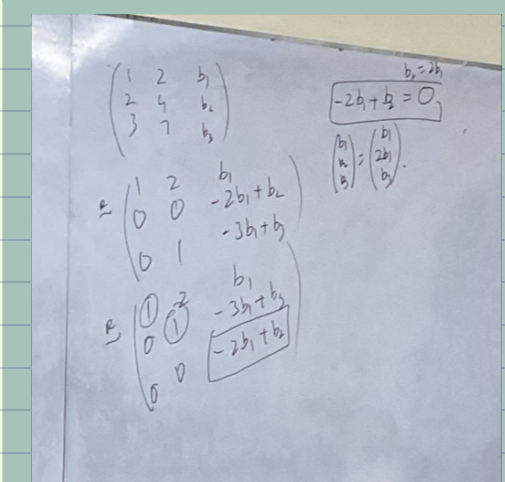
↳ whole $\mathbb{R}^2$ ⇒ $\left\{\begin{pmatrix}x\\y\end{pmatrix} = x, y \in \mathbb{R}\right\}$

→ $\text{Span}\left\{\begin{pmatrix}1\\2\end{pmatrix}, \begin{pmatrix}3\\4\end{pmatrix}, \begin{pmatrix}5\\6\end{pmatrix}\right\} \Rightarrow \mathbb{R}^2$

↓ non-parallel.

# Span of vectors in $\mathbb{R}^3$.-



$x + 2y = 4$
$2x + 4y = 8$

$x + 2y = 5$
$2x + 4y = 9$

↳ inconsistent.

$x\begin{pmatrix}1\\2\end{pmatrix} + y\begin{pmatrix}2\\4\end{pmatrix} = \begin{pmatrix}4\\8\end{pmatrix}$

L.C of $\begin{pmatrix}1\\2\end{pmatrix}\begin{pmatrix}2\\4\end{pmatrix}$

Span $\left\{\begin{pmatrix}1\\2\end{pmatrix}\begin{pmatrix}2\\4\end{pmatrix}\right\}$ → line

if $\begin{pmatrix}4\\8\end{pmatrix}$ multiple → consistent

not multiple → inconsistent

$x + 2y = 7$
$2x + 4y = 8$
$3x + 7y = 9$

$x\begin{pmatrix}1\\2\\3\end{pmatrix} + y\begin{pmatrix}2\\4\\7\end{pmatrix} = \begin{pmatrix}7\\8\\9\end{pmatrix}$ can be any value

Span $\left\{\begin{pmatrix}1\\2\\3\end{pmatrix}\begin{pmatrix}2\\4\\7\end{pmatrix}\right\}$ $\begin{pmatrix}b_1\\b_2\\b_3\end{pmatrix}$

$$\text{Span}\left\{\begin{pmatrix}1\\2\\3\end{pmatrix},\begin{pmatrix}4\\5\\6\end{pmatrix}\right\}$$

$$c_1\begin{pmatrix}1\\2\\3\end{pmatrix}+c_2\begin{pmatrix}4\\5\\6\end{pmatrix}=\begin{pmatrix}x\\y\\z\end{pmatrix}$$

$$\begin{bmatrix}1&4&x\\2&5&y\\3&6&z\end{bmatrix}$$

$$R\begin{bmatrix}1&4&x\\0&-3&2x+y\\0&-6&-3x+z\end{bmatrix}$$

$$R\begin{bmatrix}1&4&x\\0&-3&2x+y\\0&0&x-2y+z\end{bmatrix}$$

→ consistent iff $x-2y+z=0$

if it is pivot column → then inconsistent

→ should be 0

$$\text{Span}\left\{\begin{pmatrix}1\\2\\3\end{pmatrix},\begin{pmatrix}4\\5\\6\end{pmatrix}\right\}=\left\{\begin{pmatrix}x\\y\\z\end{pmatrix}: x-2y+z=0\right\}$$

Since, vectors are in $\mathbb{R}^3$, their span is in $\mathbb{R}^3$

⇒ make linear combination
⇒ the augmented matrix
⇒ reduce to echelon form
⇒ pivot column find.
⇒ if pivot → inconsistent
⇒ if not pivot → consistent.

Span $\left\{\begin{pmatrix}1\\2\end{pmatrix}, \begin{pmatrix}3\\4\end{pmatrix}\right\}$

$$c_1\begin{pmatrix}1\\2\end{pmatrix} + c_2\begin{pmatrix}3\\4\end{pmatrix} = \begin{pmatrix}x\\y\end{pmatrix}$$

$$\begin{bmatrix}1 & 3 & x\\ 2 & 4 & y\end{bmatrix}$$

$$R \begin{bmatrix}1 & 3 & x\\ 0 & -2 & -2x+y\end{bmatrix}$$

⇒ 2 vector always generate $\mathbb{R}^2$ (whole) in $\mathbb{R}^2$

⇒ but in space, 2 vector will always make a plane. whole space cannot be generated.

⇒ 3 vectors can generate whole of $\mathbb{R}_3$

System is always consistent:-

$$\begin{pmatrix} * & \checkmark \\ 0 & * \\ 0 & 0 \end{pmatrix}$$ → always pivot for some values

⇒ $\mathbb{R}_2 \longrightarrow$ 2 vector (non-multiple)

⇒ $\mathbb{R}_3 \longrightarrow$ 3

⇒ ⋮

$n \longrightarrow$ n vectors (non-multiple)

Span $\left\{ \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix}, \begin{pmatrix} 5 \\ 6 \\ 7 \\ 8 \end{pmatrix}, \begin{pmatrix} 1 \\ 3 \\ 1 \\ 5 \end{pmatrix} \right\}$

↳ cannot generate whole of $\mathbb{R}^4$

↳ because they are 3 vectors.

$$\begin{pmatrix} * & \checkmark & \checkmark \\ 0 & * & \checkmark \\ 0 & 0 & * \end{pmatrix} \longrightarrow$$ consistent.

$$\begin{pmatrix} * & \checkmark & \checkmark \\ 0 & 0 & * \\ 0 & 0 & 0 \end{pmatrix}$$ ↗ multiple of eachother

→ cannot generate $\mathbb{R}^2$

## Linearly independent Set:-

$\{v_1, v_2, v_3\}$ → linear dependent

→ one vector laying in span of other

$\left\{\begin{pmatrix}1\\2\end{pmatrix}, \begin{pmatrix}2\\4\end{pmatrix}\right\}$ → linear dependent.

$v_i = c_1v_1 + c_2v_2 + \dots c_{i-1}v_{i-1} + c_{i+1}v_{i+1} + \dots + c_nv_n$

$\{v_1, v_2, \dots, v_n\}$

$c_1v_1 + c_2v_2 + \dots c_nv_n = 0$ → homogenous Eq

↓

consistent

→ because always passes through origin

- unique Sol → zero is only Solution → trivial → linear independent
- Infinite Sol → zero + non-zero Sol → non-trivial → linear dependent

$c_1v_1 + c_2v_2 = v_3$

$c_1v_1 + c_2v_2 - v_3 = 0$

$\begin{pmatrix}c_1\\c_2\\-1\end{pmatrix}$ → became dependent

$$\left\{\begin{pmatrix}1\\2\\3\end{pmatrix},\begin{pmatrix}4\\5\\6\end{pmatrix},\begin{pmatrix}7\\8\\9\end{pmatrix}\right\}$$

$$c_1\begin{pmatrix}1\\2\\3\end{pmatrix}+c_2\begin{pmatrix}4\\5\\6\end{pmatrix}+c_3\begin{pmatrix}7\\8\\9\end{pmatrix}=\begin{pmatrix}0\\0\\0\end{pmatrix}$$

- → echelon
- → all pivot cols
- → no free var
- → 1 Sol
- → L. independent
- → trivial

- → echelon
- → not all pivot cols
- → has free var
- → infinite Sol
- → linearly dep
- → non-trivial

$$A=\begin{pmatrix}1&3\\2&4\\3&5\end{pmatrix}$$

Find span of columns of A

$$\left\{c_1\begin{pmatrix}1\\2\\3\end{pmatrix}+c_2\begin{pmatrix}3\\4\\5\end{pmatrix}=\begin{pmatrix}x\\y\\z\end{pmatrix}\right\}$$

features

$$Ax = \begin{pmatrix} v_1 & v_2 & \cdots & v_n \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$$

weights assign to each feature

regression = Ax (prediction)

vector equations tells all the weights assign to a feature

$Ax = b$ →

$Ax = 0$ → homogenous

→ 0 + span

→ span

↳ line

↓

1 free variable

translation of span

a point + span

$$\left\{ \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 3 \\ 4 \end{pmatrix}, \begin{pmatrix} 5 \\ 6 \end{pmatrix} \right\}$$

These 2 generate whole $\mathbb{R}^2$

This lies in it.

# (Vector Space)

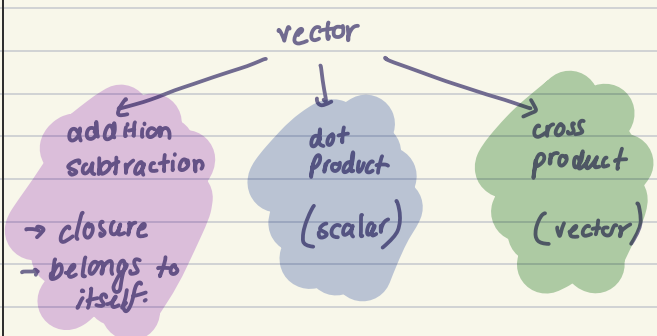
$Z_2 = \{0,1\}$
$Z_3 = \{0,1,2\}$
↳ modular classes

→ field

'V' is said to be a vector space

* if 2 operations are defined:-
  * addition of the elements
    $(V \times V) \rightarrow V$
    (out come should belong)
  * multiplication with a scalar value
    $(F \times V) \rightarrow V$

Example of Vector Space:-
- → $R^n \rightarrow R^2, R^3, R^1$
- → polynomial
- → set of all (n×n) matrics ⟹ ask (2×3)
- → functions



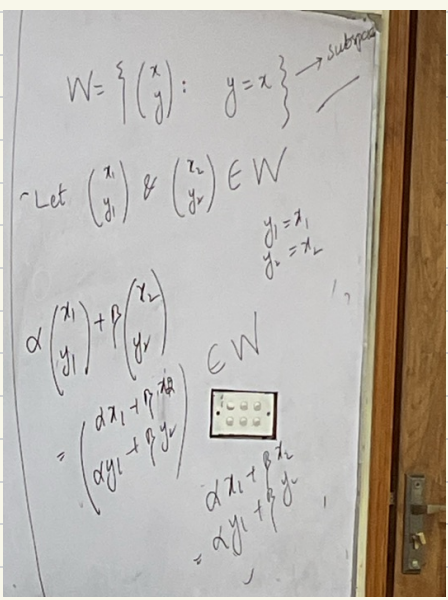
# Subspace

* $w_1 + w_2 \in W$
* $\alpha w_1 \in W$
* $0 \in W$
* $-w \in W$

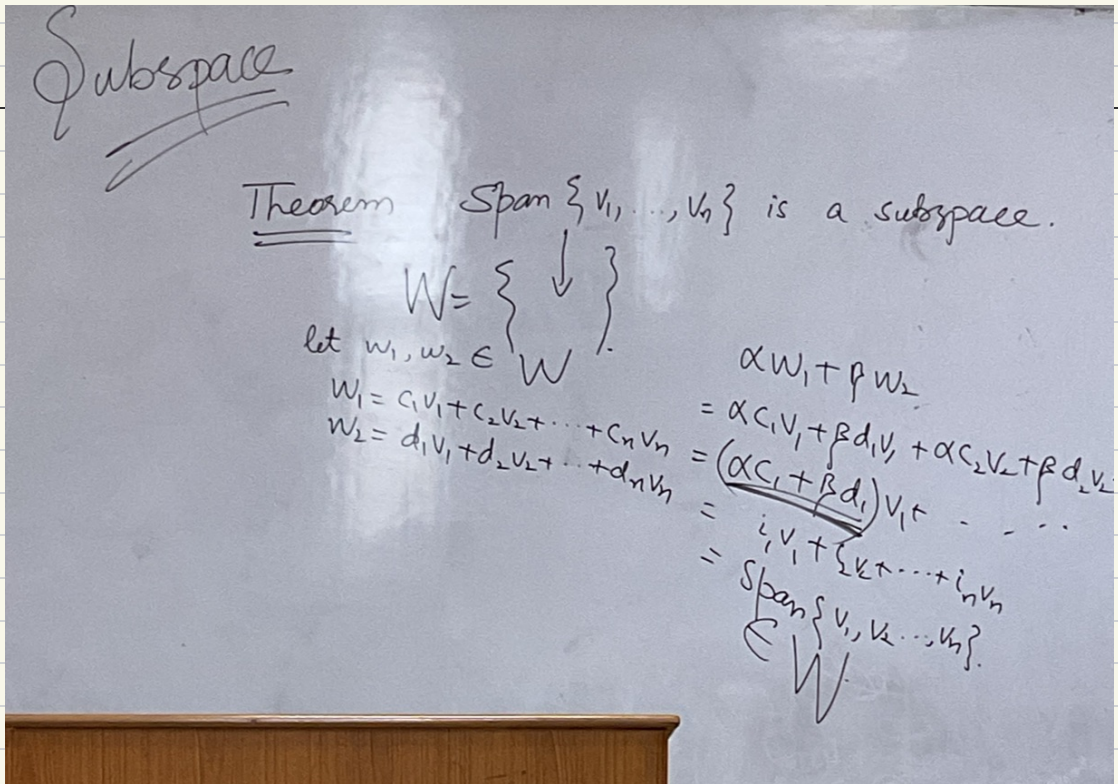
$\rightarrow \alpha w_1 + \beta w_2 \in W$

$\forall\ w_1, w_2 \in W$

$\alpha, \beta \in F(\mathbb{R})$




Span $\{v_1, \ldots v_n\}$ is a Subspace




→ Properties:-

(addition)

→ associative
→ commutative
→ identity
→ inverse

(multiplication)

→ $\alpha(u+v) = \alpha u + \alpha v$

$\alpha\beta(u) = (\alpha\beta)u$

$1 \cdot u = u$

→ finite dimensional spaces
- → $\mathbb{R}\dots$
- → matrices
- → polynomial

$W \subseteq V$

$\Rightarrow$ W is a workspace if $\alpha w_1 + \beta w_2 \in W$ $\quad \forall\ w_1, w_2 \in W$, $\alpha, \beta \in F$

* span $\{v_1, \dots, v_n\}$ is a subspace

* subset of vector space, is itself a vector space

$\mathbb{R}^2$ → Line

$\mathbb{R}^3$ → Line, Plane

(Four Fundamental Subspaces)

* Column Space of A
* Row Space of A
* Null Space of A
* Null Space of $A^t$ (Left Nullspace)

* Column Space (A)

$$A = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}$$

span of columns

* RowSpace (A)

span of rows

* NullSpace

Solution of $Ax = 0$ (homogeneous eqs)

* Left NullSpace

Solution of $A^t x = 0$

$$A = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}$$

$$Col(A) = Span\left\{\begin{pmatrix}1\\2\\3\end{pmatrix}, \begin{pmatrix}4\\5\\6\end{pmatrix}\right\} = \left\{\begin{pmatrix}x\\y\\z\end{pmatrix} : x - 2y + z = 0\right\}$$

$$C_1\begin{pmatrix}1\\2\\3\end{pmatrix} + C_2\begin{pmatrix}4\\5\\6\end{pmatrix} = \begin{pmatrix}x\\y\\z\end{pmatrix}$$

Reduce in Echelon form

$\Rightarrow$

$$A = \begin{pmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{pmatrix}$$

* A set $B$ is said to be basis of $V$

if * $\text{span}(B) = V$
* $B$ is linearly Indep

⇒ To be a basis, $R^2$ → 2 vector
$R^3$ → 3 vectors
$R^4$ → 4 vectors

Example of basis:- → $\left\{\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}\right\}$ (line)

→ $\left\{\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}\right\}$ (plane)

→ $\left\{\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}, \begin{pmatrix} 7 \\ 8 \\ 9 \end{pmatrix}, \begin{pmatrix} 10 \\ 11 \\ 12 \end{pmatrix}\right\}$ × (always linearly dependent)

* To be the basis of $R^3$ (min/max) 3 vectors are required, the vectors should be linearly independent.

$\hat{i} + \hat{j} + \hat{k}$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$ ↗ vectors in $R^3$

$$A = \begin{pmatrix} 1 & 4 & 5 & 2 \\ 2 & 5 & 6 & 3 \\ 3 & 6 & 7 & 4 \end{pmatrix}$$

Q1 find basis of CS(A)

$$\text{Col}S(A) = \text{span}\left\{\begin{pmatrix}1\\2\\3\end{pmatrix}, \begin{pmatrix}4\\5\\6\end{pmatrix}, \begin{pmatrix}5\\6\\7\end{pmatrix}, \begin{pmatrix}2\\3\\4\end{pmatrix}\right\}$$

$$\sim \begin{bmatrix} 1 & 4 & 5 & 2 \\ 2 & 5 & 6 & 3 \\ 3 & 6 & 7 & 4 \end{bmatrix} \qquad \sim \begin{bmatrix} \textcircled{1} & 4 & 5 & 2 \\ 0 & \textcircled{-3} & -4 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Basis of $\text{Col}S(A)$ $= \left\{\begin{pmatrix}1\\2\\3\end{pmatrix}, \begin{pmatrix}4\\5\\6\end{pmatrix}\right\}$

Q2 Find basis of Row(A)

$$\text{Row}(A) = \text{span}\left\{\begin{pmatrix}1\\4\\5\\2\end{pmatrix}, \begin{pmatrix}2\\5\\6\\3\end{pmatrix}, \begin{pmatrix}3\\6\\7\\4\end{pmatrix}\right\}$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 5 & 6 & 7 \\ 2 & 3 & 4 \end{pmatrix}$$ ↘ echelon form

$$R \sim \begin{pmatrix} \textcircled{1} & 2 & 3 \\ 0 & \textcircled{-3} & -6 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \qquad \text{span} = \left\{\begin{pmatrix}1\\4\\5\\2\end{pmatrix}, \begin{pmatrix}2\\5\\6\\3\end{pmatrix}\right\}$$

Q. Find the basis of Nullspace

: Already Solved

$$Ax = 0$$

$$x_1 + 4x_2 + 5x_3 + 2x_4 = 0$$
$$-3x_2 + 4x_3 - x_4 = 0$$

$x_1 =$

$$x_2 = \frac{+4x_3 - x_4}{3}$$

$$x_2 = \frac{4}{3}x_3 - \frac{1}{3}x_4$$

$$x_1 = -4x_2 - 5x_3 - 2x_4$$

$$x_1 \begin{pmatrix} 1 \\ x_2 \\ x_4 \end{pmatrix} + x_2 \begin{pmatrix} x_3 \\ x_1 \\ \vdots \end{pmatrix}$$

basis of Nullspace.

fishy fishy killing myself

hello muneeba how are you im fine thank you and you?

ok byebye

areej kch rahi hai sir ki aik bahir hai ^^

## Two methods to Prove subspace:-

* $\alpha u + \beta v \in W$
  * $\forall \rightarrow u, v \in W$
  * $\rightarrow \alpha, \beta \in \mathbb{R}$

* $0 \in W$
* $u+v \in W$
* $\alpha u \in W$

→ everything written in span is a subspace:-

↳ because it is the area/space that covers the points

↳ therefore, any points taken lie on the span

↳ that is why it is always a subspace.

→ $Col(A) = Row(A^T)$

→ $Row(A) = Col(A^T)$

$Ax = b$

↳ not a subspace bcz, 0 vector doesn't belong to it.

→ line passing through origin → $\mathbb{R}^2$

↳ $y = mx$

→ line passing through origin

↳ $y = mx$

→ plan passing through origin

→ $\mathbb{R}^3$

Basis ⟶ optimum set (building block)

* linearly ind ⟶ if you add one more ↳ becomes dependent.
* span (B) = V ⟶

Dimensions:-

⟶ the numbers of vectors define the dimensions:-

$\left\{ \begin{pmatrix}1\\2\end{pmatrix} \begin{pmatrix}3\\4\end{pmatrix} \right\}$ ⟶ 2D.

$R = \left\{ \begin{pmatrix}x\\y\\z\end{pmatrix} : 2x + y - 3 = 0 \right\}$ ↳ passing through 0

↳ it is equation of plane $2x + y = 3$

⟶ Prove:- $\begin{pmatrix}x\\y\\z\end{pmatrix} = \begin{pmatrix}x + 0y\\0x + y\\2x + y\end{pmatrix}$

$\begin{pmatrix}x\\y\\z\end{pmatrix} = x\begin{pmatrix}1\\0\\2\end{pmatrix} + y\begin{pmatrix}0\\1\\1\end{pmatrix}$ ⟶ $\text{Span}\left\{ \begin{pmatrix}1\\0\\2\end{pmatrix}, \begin{pmatrix}0\\1\\1\end{pmatrix} \right\}$

basis ⟶ 2D

↳ Sol of homogenous Eq, Null Space

kill me please

**Example::**

$$W = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} : \; x - y + z = 0 \;\; \& \;\; 2x + y - 8z = 0 \right\}$$

$$\begin{bmatrix} 1 & -1 & 3 & 0 \\ 2 & 1 & -8 & 0 \end{bmatrix}$$

$$y = \frac{10z}{3}$$

$$x = \frac{7z}{3}$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = z \begin{pmatrix} 7/3 \\ 10/3 \\ 1 \end{pmatrix} \longrightarrow \text{Span} \left\{ \begin{pmatrix} 7/3 \\ 10/3 \\ 1 \end{pmatrix} \right\}$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = z \begin{pmatrix} 7 \\ 10 \\ 3 \end{pmatrix}$$

$$B = \left\{ \begin{pmatrix} 7 \\ 10 \\ 3 \end{pmatrix} \right\}$$

$$A = \begin{bmatrix} 1 & 3 & 7 & 11 & 1 \\ 2 & 4 & 8 & 12 & 2 \\ 3 & 5 & 9 & 13 & 3 \\ 4 & 6 & 10 & 14 & 4 \end{bmatrix}$$

→ Column → (Always original vectors in basis)

→ Rows → (can be echelon / original / inbetween)

Find the basis of the column space that are not from A.

**Row Space of $(A^T)$ = Col $(A)$**

$$A \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix} \rightarrow A^T = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$$

$$R \begin{pmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \end{pmatrix}$$

Basis of Row Space $(A^T)$ $= \left\{ \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 0 \\ -3 \\ -6 \end{pmatrix} \right\}$

↳ Column Space $(A) = \left\{ \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \begin{pmatrix} 0 \\ -3 \\ -6 \end{pmatrix} \right\}$

Find the basis of the row space that are not from A.

$$A = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}$$

$$R \begin{pmatrix} 1 & 4 \\ 0 & -3 \\ 0 & 0 \end{pmatrix}$$

$P_2$ : set of all polynomials of degree 2 and less than 2

Isomorphism

$P_2$ : $\{a_0 + a_1 x + a_2 x^2 : a_0, a_1, a_2 \in \mathbb{R}\}$

$\hookrightarrow \left\{\begin{pmatrix} a_0 \\ a_1 \\ a_2 \end{pmatrix} : a_0, a_1, a_2 \in \mathbb{R}\right\}$

$\hookrightarrow \mathbb{R}^3$

$P_2$ $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$

$\downarrow$

$1 + x_1 + x_2^2$

$P_1$ $\begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \rightarrow 1 + x$

$\mathbb{R}^2$ $\left\{\begin{pmatrix} x \\ y \end{pmatrix} : y = x\right\}$

$P_1$ $\{a_0 + a_1 x : a_0 = a_1\}$

$\mathbb{R}^3$ $= \left\{\begin{pmatrix} x \\ y \\ z \end{pmatrix} : x - y + 2z = 0\right.$

* **Polynomial:-**

$P_1 = \{ a_0 + a_1 x : a_0, a_1 \in \mathbb{R} \}$

$P_2 = \{ a_0 + a_1 x + a_2 x^2 : a_0, a_1, a_2 \in \mathbb{R} \}$

$P_3 = \{ a_0 + a_1 x + a_2 x^2 + a_3 x^3 : a_0, a_1, a_2, a_3 \in \mathbb{R} \}$

degree of polynomial 3

④ → different dimensions

The dimensions of polynomial always (1) more than degree of polynomial

**Dimension of matrix:-** = number of elements

2×2 = 4 element → 4D

2×3 = 6 elements → 6D

symmetric matrix → 2×2 → ② diagonal, ② off-diag-/2

3×3 → ③ diagonal, ⑥ off-diag../2

n×n → ⓝ diagonal → $n^2 - n$ off-diag/2

$$n + \frac{n^2 - n}{2} \Rightarrow \frac{2n + n^2 - n}{2}$$

$$\frac{n^2 + n}{2} \qquad \frac{n(n+1)}{2}$$

$$I = \{(1+x), (2+7x)\}$$

Method 1:-

$$c_1(1+x) + c_2(2+7x) = 0$$
$$c_1 + c_1x + 2c_2 + 7c_2x = 0$$
$$(c_1 + 2c_2) + (c_1x + 7c_2x) = 0$$

$$c_1 + 2c_2 = 0$$
$$c_1x + 7c_2x = 0$$

$$\begin{pmatrix} 1 & 2 \\ 1 & 7 \end{pmatrix}$$

Method 2:-

$$\left\{ \begin{pmatrix} a_0 \\ a_1 \end{pmatrix} : a_0, a_1 \in \mathbb{R} \right\}$$

$$I = \{1+x, 2+7x\}$$

$$\left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 7 \end{pmatrix} \right\}$$

Example #2

$$A = \{2+x+x^2, \quad 4+x-x^2, \quad 4+7x+x^2\}$$

$$c_1(2+x+x^2) + c_2(4+x-x^2) + c_3(4+7x+x^2) = 0$$
$$(2c_1 + 4c_2 + 4c_3) + (c_1x + c_2x + 7c_3x) + (c_1x^2 - c_2x^2 + c_3x^2) = 0$$

$$2c_1 + 4c_2 + 4c_3 = 0$$
$$c_1 x + c_2 x + 7c_3 x = 0$$
$$c_1 x^2 - c_2 x^2 + c_3 x^2 = 0$$

$$\begin{bmatrix} 2 & 4 & 4 & 0 \\ 1 & 1 & 7 & 0 \\ 1 & -1 & 1 & 0 \end{bmatrix}$$

**Method #2**

$$\begin{bmatrix} 2 & 4 & 4 \\ 1 & 1 & 7 \\ 1 & -1 & 1 \end{bmatrix}$$

directly from the polynomials.

format

$$\begin{pmatrix} a_0 \\ a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix}$$

Q Find Span $\{(1+7x)\}$

**Method 1:-**

$$c_1(1+7x) = a_0 + a_1 x$$
$$c_1 + 7c_1 x = a_0 + a_1 x$$
$$c_1 = a_0 \qquad 7c_1 = a_1$$
$$c_1 = a_1/7$$
$$a_0 = a_1/7$$

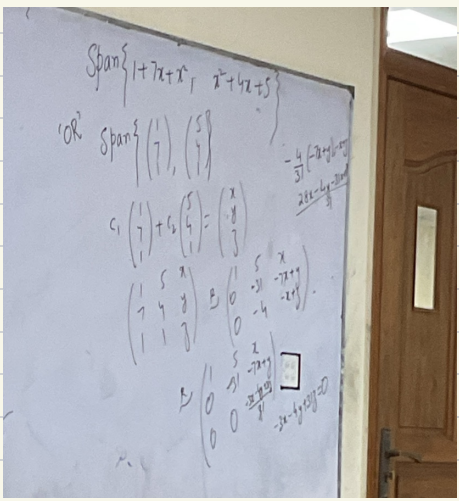
$$\text{Span}\{1+7x\} = \{a_0 + a_1 x : a_0 + \tfrac{a_1}{7}\}$$

**Method 2:-**

$$\text{Span}\left\{\begin{pmatrix} 1 \\ 7 \end{pmatrix}\right\}$$
$$y = 7x$$
$$\left\{\begin{pmatrix} x \\ y \end{pmatrix} : y = 7x\right\}$$
$$\left\{\begin{pmatrix} a_0 \\ a_1 \end{pmatrix} : a_1 = 7a_0\right\}$$
$$\{a_0 + a_1 x : a_1 = 7a_0\}$$

$\text{Span}\left\{1+7x+x^2,\ x^2+4x+5\right\}$

$\text{Span}\left\{\begin{pmatrix}1\\7\\1\end{pmatrix},\begin{pmatrix}5\\4\\1\end{pmatrix}\right\}$




$$\left\{\begin{pmatrix}x\\y\\z\end{pmatrix} : -3x-4y+31z=0\right\}$$

$$\left\{\begin{pmatrix}a_0\\a_1\\a_2\end{pmatrix} : -3a_0-4a_1+31a_2=0\right\}$$

$$\left\{a_0+a_1x+a_2x^2 : -3a_0-4a_1+31a_2\right.$$

Matrix:-

$$\left\{\begin{pmatrix}1&2\\3&4\end{pmatrix},\begin{pmatrix}5&6\\7&8\end{pmatrix},\begin{pmatrix}1&3\\4&5\end{pmatrix}\right\}$$

Method 1

$$c_1\begin{pmatrix}1&2\\3&4\end{pmatrix}+c_2\begin{pmatrix}5&6\\7&8\end{pmatrix}+c_3\begin{pmatrix}1&3\\4&5\end{pmatrix}=\begin{pmatrix}0&0\\0&0\end{pmatrix}$$

$$\begin{pmatrix}c_1+5c_2+c_3 & 2c_1+6c_2+3c_3\\ 3c_1+7c_2+4c_3 & 4c_1+8c_2+5c_3\end{pmatrix}=\begin{pmatrix}0&0\\0&0\end{pmatrix}$$

* $c_1 + 5c_2 + c_3 = 0$
* $2c_1 + 6c_2 + 3c_3 = 0$
* $3c_1 + 7c_2 + 4c_3 = 0$
* $4c_1 + 8c_2 + 5c_3 = 0$

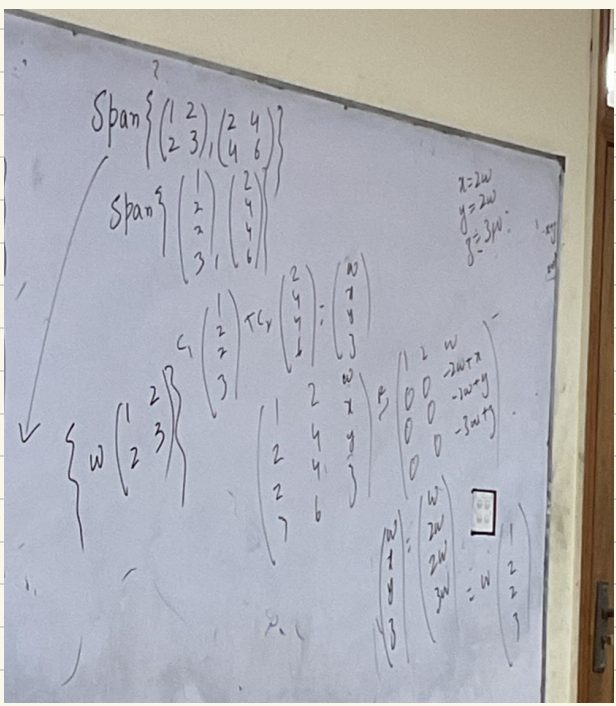
$\begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$

Method 2:-

$\begin{pmatrix} a & b \\ c & a \end{pmatrix} \sim \begin{pmatrix} a \\ b \\ c \\ a \end{pmatrix}$

$\begin{pmatrix} 1 & 5 & 1 \\ 2 & 6 & 3 \\ 3 & 7 & 4 \\ 4 & 8 & 5 \end{pmatrix}$

* Span of matrices:-



**Rank:-**

* dimension of rowspace / columnspace.
* Number of non zero rows.

**Nullity:-**

* dimension of null space → number of free variables

**Rank - Nullity Theorem:**

Rank (A) + Nullity (A) = No. of Columns.

*

$A_{5\times 3}$ → Null space is a line $R^3$

- 3 → no of column
- Null space → 1 free variable
- 2 remaining basic variables → making a plane
- where → in total number of rows which is $R_5$

dimensions → number of vectors in basis.

4 fundamental subspaces:-

* Col(A)
* Row(A)

→ basis → having pivot column

* NullSpace — $Ax = 0$
* Left NullSpace. — $A^T x = 0$

## Coordinate Vectors:-

(indirect application of LA)

→ $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$

$$\begin{pmatrix} 1 \\ 2 \end{pmatrix} = c_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + c_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix} \qquad \left[\begin{pmatrix} 1 \\ 2 \end{pmatrix}\right]_B = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

$$\begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} \Rightarrow \begin{matrix} c_1 = 1 \\ c_2 = 2 \end{matrix}$$

$I = \left\{ \begin{pmatrix} 2 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$

$$\begin{pmatrix} 2 \\ 1 \end{pmatrix} = c_1 \begin{pmatrix} 2 \\ 0 \end{pmatrix} + c_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 2c_1 \\ c_2 \end{pmatrix}$$

$c_1 = 1 \;,\; c_2 = 1$

⇒ basis means scaling → defines what scale we are using in a coordinate axis.

Generalized form:-

$X = 3b_1 + b_2$

$[x]_c = [3b_1 + b_2]_c$

$= 3[b_1]_c + 1[b_2]_c$

$[x]_c = \begin{pmatrix} [b_1]_c & [b_2]_c \end{pmatrix} \begin{pmatrix} 3 \\ 1 \end{pmatrix}$

$[x]_c = {}_{c\leftarrow B}P\,[x]_B$

Change of Basis:-

* find matrix
* multiply with C.V

$B = \{b_1, b_2\}$ $\quad C = \{c_1, c_2\}$

$b_1 = 4c_1 + c_2$ , $b_2 = -6c_1 + c_2$

$x = 3b_1 + b_2 \rightarrow [x]_B = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$

Suppose:-

$[x]_c = ?$ $\quad x = a_1c_1 + a_2c_2$

$x = 3b_1 + b_2$

$= 3(4c_1 + c_2) + (-6c_1 + c_2)$

$= 6c_1 + 4c_2$

$[x]_c = \begin{pmatrix} 6 \\ 4 \end{pmatrix}$

$b_1 = \begin{pmatrix} -9 \\ 1 \end{pmatrix}$ , $b_2 = \begin{pmatrix} -5 \\ -1 \end{pmatrix}$ $\quad c_1 = \begin{pmatrix} 1 \\ -4 \end{pmatrix}$, $c_2 \begin{pmatrix} 3 \\ -5 \end{pmatrix}$

${}_{c\leftarrow B}P$ → change of basis matrix which will transform the coordinate vector w.r.t B to coordinate vector w.r.t C.

${}_{c\leftarrow B}P = \begin{pmatrix} [b_1]_c & [b_2]_c \end{pmatrix}$

$$b_1 = a_1\begin{pmatrix}1\\-4\end{pmatrix} + a_2\begin{pmatrix}3\\-5\end{pmatrix}$$

$$b_2 = d_1\begin{pmatrix}1\\-4\end{pmatrix} + d_2\begin{pmatrix}3\\-5\end{pmatrix}$$

$$[b_1]_C = \begin{pmatrix}6\\-5\end{pmatrix}, \quad [b_2]_C = \begin{pmatrix}4\\-3\end{pmatrix}$$

$$P_{C\leftarrow B} = \begin{pmatrix}6 & 4\\-5 & -3\end{pmatrix}$$

$$a_1\begin{pmatrix}1\\-4\end{pmatrix} + a_2\begin{pmatrix}3\\-5\end{pmatrix} = \begin{pmatrix}-9\\1\end{pmatrix} \Rightarrow Ax = b_1$$

$$d_1\begin{pmatrix}1\\-4\end{pmatrix} + d_2\begin{pmatrix}3\\-5\end{pmatrix} = \begin{pmatrix}-5\\-1\end{pmatrix} \Rightarrow Ay = b_2$$

$$(A \;\; b_1), (A \;\; b_2)$$

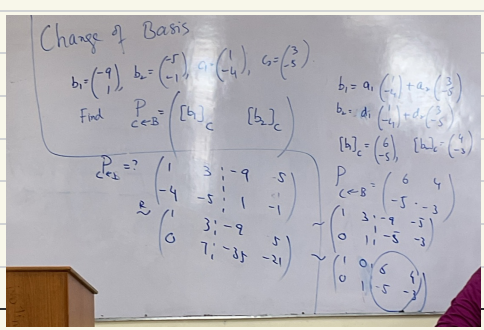
$$\left(\begin{array}{cc:cc} & & & \\ & A & b_1 & b_2 \\ & & & \end{array}\right)$$

$$P_{C\leftarrow B} = \left(\begin{array}{cc:cc} 1 & 3 & -9 & 5 \\ -4 & -5 & 1 & -1 \end{array}\right)$$

$$\left(\begin{array}{cc|cc} | & | & | & | \\ c_1 & c_2 & b_1 & b_2 \\ | & | & | & | \end{array}\right)$$

$$\left(\begin{array}{cc|cc} & & a_1 & d_1 \\ c_1 & c_2 & & \\ & & a_2 & d_2 \end{array}\right)$$

$$\left(\begin{array}{cc|c} 1 & 0 & \\ 0 & 1 & P_{C\leftarrow B} \end{array}\right)$$



1.1 → 1.5, 1.7, 4.1 – 4.7

## Linear Transformation:-

$f : \mathbb{R} \longrightarrow \mathbb{R}$ (real values)

$f : \mathbb{R}^2 \longrightarrow \mathbb{R}$ (real values)

$f : \mathbb{R}^n \rightarrow \mathbb{R}^m$

→ domain → input

→ co-domain → candidate set from which output comes

→ range → output.

→ matrix transformation:-

$T(u) = Ax$

matrix of transformation

Generalized:-

$T : \mathbb{R}^n \rightarrow \mathbb{R}^m$

T: vector from $\mathbb{R}^n$; gives vector in $\mathbb{R}^m$

$T : V \longrightarrow W$

V: vector space; W: vector space

→ polynomial → matrices

→ matrix → real number → determinant

### Conditions

$T(v+w) = T(v) + T(w)$

$T(cu) = cT(u)$

real → from set of domain V

transformation → from set of domain of W

$T(u+y) = A(u+y)$

$= A(u) + A(y)$

$= T(u) + T(y)$

$T(cx) = A(cx)$

$= cA$

$T(cx) = cT(u)$

$T: \mathbb{R}^2 \to \mathbb{R}^2$

$$T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x+y \\ 2x+9y \end{pmatrix}$$

$$\mathbb{R}^2 = \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = c_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + c_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$c_1 = x$ , $c_2 = y$

$$\begin{pmatrix} x \\ y \end{pmatrix} = x \begin{pmatrix} 1 \\ 0 \end{pmatrix} + y \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$T\begin{pmatrix} x \\ y \end{pmatrix} = T\left[ x \begin{pmatrix} 1 \\ 0 \end{pmatrix} + y \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right]$$

$$T\begin{pmatrix} x \\ y \end{pmatrix} = x\, T\begin{pmatrix} 1 \\ 0 \end{pmatrix} + y\, T\begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$T\begin{pmatrix} x \\ y \end{pmatrix} = \left[ T\begin{pmatrix} 1 \\ 0 \end{pmatrix} \; T\begin{pmatrix} 0 \\ 1 \end{pmatrix} \right] \begin{bmatrix} x \\ y \end{bmatrix}$$

$T(x) = Ax$

→ $T: \mathbb{R}^2 \to \mathbb{R}$

$$B = \left\{ \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 3 \\ 4 \end{pmatrix} \right\}$$

$$c_1 \begin{pmatrix} 1 \\ 2 \end{pmatrix} + c_2 \begin{pmatrix} 3 \\ 4 \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\begin{pmatrix} 1 & 3 & x \\ 2 & 4 & y \end{pmatrix}$$

* diagonal
* triangular

$R_2$

$$\begin{pmatrix} 1 & 3 & x \\ 0 & -2 & 2x-y \end{pmatrix}$$

$$c_2 = \frac{2x-y}{-2}$$

$$c_1 = \frac{-4x+3y}{2}$$

$$T\begin{pmatrix} x \\ y \end{pmatrix} = \frac{-4x+3y}{2} T\begin{pmatrix} 1 \\ 2 \end{pmatrix} + \left(\frac{2x-y}{-2}\right) T\begin{pmatrix} 3 \\ 4 \end{pmatrix}$$

---

## (Sessional -2)

**Recap:-**

Linear Transformation:-

* a line will map a line to line → never adds bends/curve
* it never moves the origin (translates)

$$T_C(u) = CT(u)$$
$$T_C(0) = CT(0)$$
$$0 = 0$$

$T(x) = AX$

$T(x) = AX$

$$T\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

$$T\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} x_1 + 3x_2 \\ 2x_1 + 4x_2 \end{pmatrix}$$

$T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$

$$T\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

$$T\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} x_1 + 4x_2 \\ 2x_1 + 5x_2 \\ 3x_1 + 6x_2 \end{pmatrix}$$

$T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$

dear count = 0

add count, 3

Kernel methods

$$T\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

$$T\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} x_1 + 3x_2 + 5x_3 \\ 2x_1 + 4x_2 + 6x_3 \end{pmatrix}$$

$T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$

$A_{m \times n}$

n vectors from $\mathbb{R}^n$

$T(x) = A_{mn} X_{n \times 1}$

$T: \mathbb{R}^n \rightarrow \mathbb{R}^m$

* we are taking n vectors
* they need n scalars
* so the transformation is from $\mathbb{R}^n$
* there are m elements in the transforming matrix.
* result is in $\mathbb{R}^m$

$A_{5 \times 6}$

$T: \mathbb{R}^6 \rightarrow \mathbb{R}^5$

Q:- Why do we study linear transformation

# Linear transformation in Computer Vision

(Application of linear transformation)

* contraction
* scaling
* rotation
* reflection

→ from $\mathbb{R}_2 \rightarrow \mathbb{R}_2$

$$T\begin{pmatrix} x \\ y \end{pmatrix} = A\begin{pmatrix} x \\ y \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = c_1\begin{pmatrix} 1 \\ 0 \end{pmatrix} + c_2\begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = x\begin{pmatrix} 1 \\ 0 \end{pmatrix} + y\begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$T\begin{pmatrix} x \\ y \end{pmatrix} = T\left[x\begin{pmatrix} 1 \\ 0 \end{pmatrix} + y\begin{pmatrix} 0 \\ 1 \end{pmatrix}\right]$$

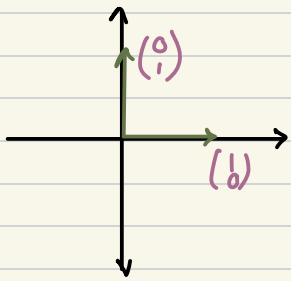
$$= T\left[x\begin{pmatrix} 1 \\ 0 \end{pmatrix}\right] + T\left[y\begin{pmatrix} 0 \\ 1 \end{pmatrix}\right]$$

$$= x\,T\begin{pmatrix} 1 \\ 0 \end{pmatrix} + y\,T\begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$T\begin{pmatrix} x \\ y \end{pmatrix} = \left(T\begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad T\begin{pmatrix} 0 \\ 1 \end{pmatrix}\right)\begin{pmatrix} x \\ y \end{pmatrix}$$

(flip)

⇒ Reflection about $x$-axis

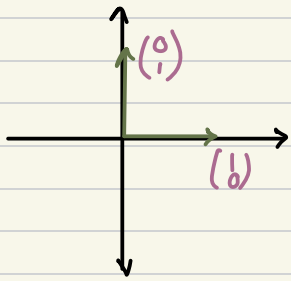



$$T\begin{pmatrix}1\\0\end{pmatrix} = \begin{pmatrix}1\\0\end{pmatrix}$$

$$T\begin{pmatrix}0\\1\end{pmatrix} = \begin{pmatrix}0\\-1\end{pmatrix}$$

$$A = \begin{pmatrix}1 & 0\\0 & -1\end{pmatrix}$$

Reflection about $x$-axis




$$T\begin{pmatrix}1\\0\end{pmatrix} = \begin{pmatrix}-1\\0\end{pmatrix}$$

$$T\begin{pmatrix}0\\1\end{pmatrix} = \begin{pmatrix}0\\1\end{pmatrix}$$

$$A = \begin{pmatrix}-1 & 0\\0 & 1\end{pmatrix}$$

⇒ Reflection Origin:-

$$T\begin{pmatrix}1\\0\end{pmatrix} = \begin{pmatrix}-1\\0\end{pmatrix}$$

$$T\begin{pmatrix}0\\1\end{pmatrix} = \begin{pmatrix}0\\-1\end{pmatrix}$$

$$\begin{pmatrix}-1 & 0\\0 & -1\end{pmatrix}$$

Countttt

## Contraction

$$T\begin{pmatrix}1\\0\end{pmatrix} = \begin{pmatrix}1/2\\0\end{pmatrix}$$

$$T\begin{pmatrix}0\\1\end{pmatrix} = \begin{pmatrix}0\\1/2\end{pmatrix}$$

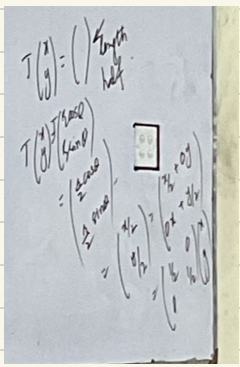
$$A = \begin{pmatrix}1/2 & 0\\0 & 1/2\end{pmatrix}$$

## Dilation

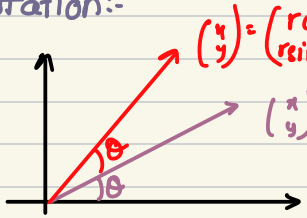
$$T\begin{pmatrix}1\\0\end{pmatrix} = \begin{pmatrix}2\\0\end{pmatrix}$$

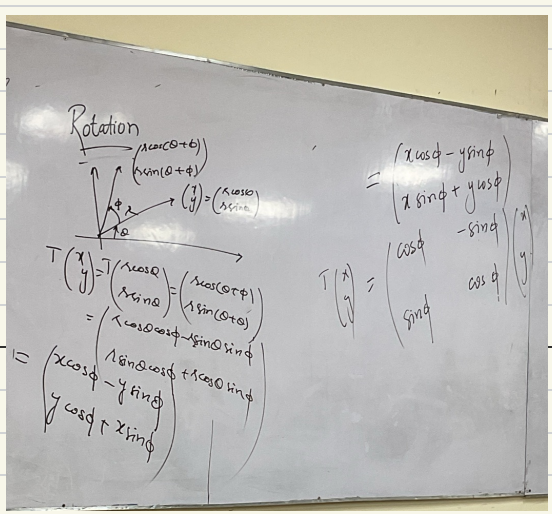
$$T\begin{pmatrix}0\\1\end{pmatrix} = \begin{pmatrix}0\\2\end{pmatrix}$$

$$A = \begin{pmatrix}2 & 0\\0 & 2\end{pmatrix}$$



## Rotation:-



Method 2:-

$$T\begin{pmatrix}1\\0\end{pmatrix} = \begin{pmatrix}\cos\phi\\ \sin\phi\end{pmatrix} \qquad \begin{pmatrix}C & -S\\ S & C\end{pmatrix}$$

$$T\begin{pmatrix}0\\1\end{pmatrix} = \begin{pmatrix}\cos(\pi/2+\phi)\\ \sin(\pi/2+\phi)\end{pmatrix}$$

$$= \begin{pmatrix}-\sin\phi\\ \cos\phi\end{pmatrix}$$

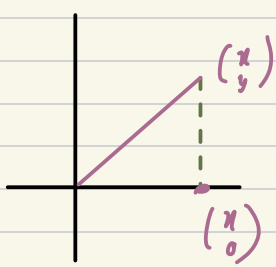
Matrix of Linear Transformation

Projection:-

proj on $x$-axis

$$T\begin{pmatrix}1\\0\end{pmatrix} = \begin{pmatrix}1\\0\end{pmatrix} \qquad T\begin{pmatrix}0\\1\end{pmatrix} = \begin{pmatrix}0\\0\end{pmatrix}$$

$$A = \begin{pmatrix}1 & 0\\ 0 & 0\end{pmatrix}$$




$$T\begin{pmatrix}x\\y\end{pmatrix} = \begin{pmatrix}x\\0\end{pmatrix}$$

$$= \begin{pmatrix}1 & 0\\ 0 & 0\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix}$$

Reflection about xy plane:-

$$T\begin{pmatrix}1\\0\\0\end{pmatrix}=\begin{pmatrix}1\\0\\0\end{pmatrix}$$

$$T\begin{pmatrix}0\\1\\0\end{pmatrix}=\begin{pmatrix}0\\1\\0\end{pmatrix}$$

$$T\begin{pmatrix}0\\0\\1\end{pmatrix}=\begin{pmatrix}0\\0\\-1\end{pmatrix}$$

$$T\begin{pmatrix}x\\y\\z\end{pmatrix}=\begin{pmatrix}x\\y\\-z\end{pmatrix}$$

$$T\begin{pmatrix}x\\y\\z\end{pmatrix}=\begin{pmatrix}1&0&0\\0&1&0\\0&0&-1\end{pmatrix}\begin{pmatrix}x\\y\\z\end{pmatrix}$$

⇒ Linear transformation:-

Q#1 $T: P_2 \to P_1 \sim \{1, x, x^2\}$

$T(p(x)) = p'(x)$

$P_2 \to$ 3 dimensions

$P_1 \to$ 2 dimensions

$T(1) = 0$

$T(x) = 1$

$T(x^2) = 2x$

$$[0]_c = \begin{pmatrix}0\\0\end{pmatrix} \qquad 0 = a_0 + a_1 x \quad a_0 = 0,\ a_1 = 0$$

$$[1]_c = \begin{pmatrix}1\\0\end{pmatrix} \qquad 1 = a_0 + a_1 x \quad a_0 = 1,\ a_1 = 0$$

$$[2x]_c = \begin{pmatrix}0\\2\end{pmatrix} \qquad 2x = a_0 + a_1 x \quad a_0 = 0,\ a_1 = 2$$

$$A = \begin{pmatrix}0&1&0\\0&0&2\end{pmatrix}$$

## Q#2

$T: \mathbb{R}^2 \to P_1$

$T\begin{pmatrix} a_0 \\ a_1 \end{pmatrix} = a_0 + a_1 x$

$T\begin{pmatrix} 1 \\ 0 \end{pmatrix} = 1 + 0x$

$T\begin{pmatrix} 0 \\ 1 \end{pmatrix} = 0 + 1x$

$[1]_C = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

$[x]_C = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

$A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

$A \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & -1 \\ 1 & 1 \end{pmatrix}$

## Q#3

$T: P_1 \to (M_{2\times 2})$

$T(a_0 + a_1 x) = \begin{pmatrix} a_0 & a_1 \\ a_0 - a_1 & a_0 + a_1 \end{pmatrix}$

$a_0 = 1 \quad a_1 = 0$

$T(1) = T(1 + 0x) = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$

$a_0 = 0 \quad a_1 = 1$

$T(x) = T(0 + 1x) = \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix}$

Standard basis:-

$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$

$\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} = c_1 \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + c_2 \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} + c_3 \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} + c_4 \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$

by comparison:-

$c_1 = 1, \quad c_2 = 0, \quad c_3 = 1, \quad c_4 = 1$

$\begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix} = c_1 \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + c_2 \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} + c_3 \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} + c_4 \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$

by comparison:-

$c_1 = 0, \quad c_1 = 1, \quad c_2 = -1, \quad c_3 = 1$

first take:-

* domain basis (standard ones)
* perform transformation
* then write coordinate vectors
* compare values
* write matrix

→ $\left( \left[ T\begin{pmatrix}1\\0\end{pmatrix} \right]_C \left[ T\begin{pmatrix}0\\1\end{pmatrix} \right]_C \right)$ → not a standard basis.

① how to write transformation
② given T → write matrix → (multiply with cordinate)

implement

If we multiply any vector with this transformation, we will get its derivative

$T: P_2 \to P_1 \quad T(p) = p'$

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

$T(2 + 7x + 9x^2)$

↳ cordinate vector

$[2 + 7x + 9x^2]_C$

$T(x) = A(x)$

$$T\begin{pmatrix}2\\7\\9\end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}\begin{pmatrix}2\\7\\9\end{pmatrix}$$

$$= \begin{pmatrix}7\\18\end{pmatrix}$$

$= (7 + 18x)$

# Range of Transformation:-

→ column space ⟶ range of transformation

$T: \mathbb{R}^2 \to \mathbb{R}^2$

$$T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x-y \\ 2x+y \end{pmatrix}$$

$$= \begin{pmatrix} 1 & -1 \\ 2 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}$$

$$= x\begin{pmatrix} 1 \\ 2 \end{pmatrix} + y\begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

$$T\begin{pmatrix} x \\ y \end{pmatrix} = x\begin{pmatrix} 1 \\ 2 \end{pmatrix} + y\begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

$$= \text{l.c of } \begin{pmatrix} 1 \\ 2 \end{pmatrix} \& \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

$$= \text{Span}\left\{\begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \end{pmatrix}\right\}$$

↓

whole of $\mathbb{R}^2$

$$T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x+2y \\ 2x+4y \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}$$

↓

line $y = 2x$

↙

$\begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ 4 \end{pmatrix}, \begin{pmatrix} 3 \\ 6 \end{pmatrix}$

## Kernal of L.T

⟶ kernal is related to null space

$T(x) = Ax$

$T(x) = 0$

$A(x) = 0$

↳ null space.

# Composition of Linear Transformation

$T: U \to V$ (same) $\quad S: W \to U$ (same)

$x \in W$

$T(x) = Ax \quad S(x) = Bx \quad S(x) \in Y = U$

$$T \circ S(x) = T(S(x)) = T(Bx) = ABx$$

$A^2 = I$

$AA = I$

1st transformation

2nd tranformation

no effect

eg. rotate about $\pi$

$T: U \to W \quad S: W \to V$

$T(x) = Ax, \quad S(x) = Bx$

$$S \circ T(x) = S(T(x)) = S(Ax) = BAx$$

$AB = 0$

projection about $x$ $\to$ $y = 0$

projection about $y$ $\to$ $x = 0$

Ref — x-axis, y-axis, origin, rotation } $R^2$

Ref — xy-plane, yz-plane, zx-plane } $R_3$

Rotation:-

→ x-axis
→ y-axis
→ z-axis

Ref x-axis , Rotate $\pi/2$ , Ref y-axis

method 1. (unrecommended)

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$$

$A_1$ $A_2$ $A_3$

$T_1$ $T_2$ $T_3$

$$(T_3 \circ (T_2 \circ (T_1)))$$

↳ $A_3 A_2 A_1$

Method 2:-

$$\begin{pmatrix}1\\0\end{pmatrix} \xrightarrow{\text{ref } x} \begin{pmatrix}1\\0\end{pmatrix} \xrightarrow{\text{rotate } \pi/2} \begin{pmatrix}0\\1\end{pmatrix} \xrightarrow{\text{ref } y} \begin{pmatrix}0\\1\end{pmatrix}$$

$$\begin{pmatrix}0\\1\end{pmatrix} \xrightarrow[\text{ref } x]{} \begin{pmatrix}0\\-1\end{pmatrix} \xrightarrow[\pi/2]{\text{rotate}} \begin{pmatrix}1\\0\end{pmatrix} \xrightarrow{\text{ref } y} \begin{pmatrix}-1\\0\end{pmatrix}$$

## Homogenous Coordinates:-

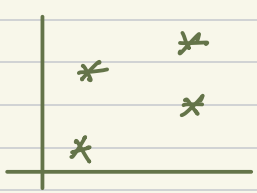
$\begin{pmatrix} x \\ y \end{pmatrix} \rightarrow \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}$ → to incoperate translations:-

$$T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x+h \\ y+k \end{pmatrix}$$

eg $T\begin{pmatrix} 0 \\ 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 \\ 7 \end{pmatrix}$ $\quad T\begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 5 \\ 9 \end{pmatrix}$

$$\begin{pmatrix} 1 & 0 & h \\ 0 & 1 & k \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \\ 1 \end{pmatrix} = \begin{pmatrix} x+h \\ y+k \\ 1 \end{pmatrix}$$

→ mapping :-



2 horizontal
3 vertical

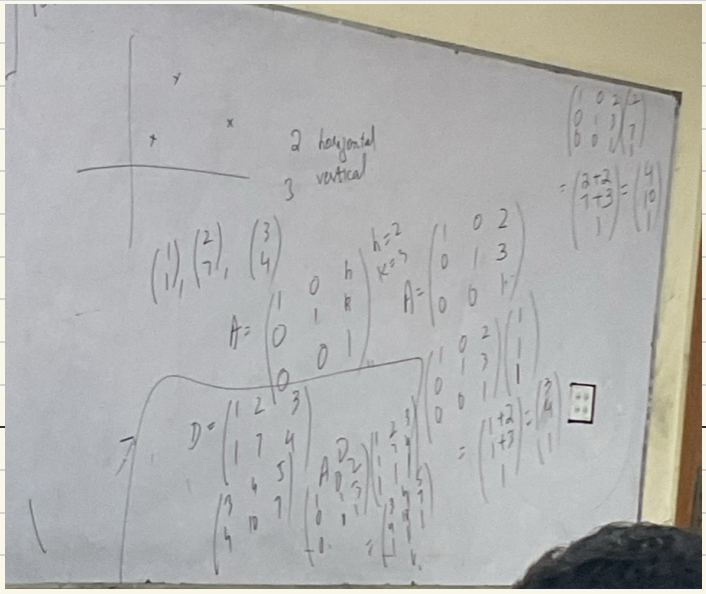
$$A \begin{pmatrix} 1 & 0 & h \\ 0 & 1 & k \\ 0 & 0 & 1 \end{pmatrix} \rightarrow A \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{pmatrix}$$

multiply vectors to translate them.

$$\begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$$

Homogeneous

or instead MUL the whole matrix



rotate $\pi/2$, translate $h=1, h=2$

$$A_1 \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad A_2 = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix}$$

Why happened

$$\begin{pmatrix} c & -s \\ s & c \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} cx - sy \\ sx + cy \end{pmatrix}$$

$$\begin{pmatrix} c & -s & 0 \\ s & c & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} \quad \begin{pmatrix} cx - sy \\ sx + cy \\ 1 \end{pmatrix}$$

## Inverse of matrix:-

↳ transformation inverse

**method 1:-**

$$A^{-1} = \frac{Adj(A)}{|A|}$$

**method 2:-**

$A^{-1} = ?$

$$\left( A \vdots I \right)$$

$$\left( I : A^{-1} \right)$$

**method 2:-**

$$A\begin{pmatrix}1 & 2\\ 3 & 4\end{pmatrix}$$

$$\left(\begin{array}{cc:cc}1 & 2 & 1 & 0\\ 3 & 4 & 0 & 1\end{array}\right)$$

$$\left(\begin{array}{cc|cc}1 & 2 & 1 & 0\\ 0 & 1 & 3/2 & -1/2\end{array}\right)$$

$$\left(\begin{array}{cc|cc}1 & 0 & -2 & 1\\ 0 & 1 & 3/2 & -1/2\end{array}\right)$$

$Ax = e_1$ ↳ $x = A^{-1}(e_1)$

$Ay = e_2$ ↳ $y = A^{-1}(e_2)$

$B(e_1) = B\begin{pmatrix}1\\0\end{pmatrix}$ → 1st column of B.

$A^{-1}_{2\times 2}$ → 2 columns

1st Col ← $A^{-1}e_1$ $\quad$ $A^{-1}e_2$ → 2nd Col

$$Ax = e_1$$

$$\left(\begin{array}{cc:cc}1 & 3 & 1 & 0\\ 2 & 4 & 0 & 1\end{array}\right)$$

# Inverse of Linear Transformation:-

$f: A \to B$

→ one - one $\{$ if $x_1 \neq x_2$, then $f(x_1) \neq f(x_2)\}$

eg $f(x) = \sin(x)$

$f(0) = 0$

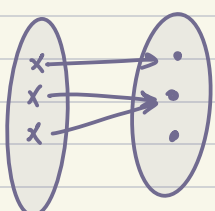
$f(\pi) = 0$ not one - one

→ onto

$\{\forall\ y_1 \in B\ \exists\ x_1 \geq A$

$f(x_1) = y_1 \Rightarrow$ (Codomain = Range)$\}$

→ Bijective (invertible)



(not one-one)

T. $U \to V$

→ one - one

→ onto

$\}$ → invertible

dear count = 1

One - one LT:-

'A' ⟶ echelon form

→ if each column has the pivot

→ then 1-1

$T(x) = Ax$

$T(x_1) = Ax_1$

$T(x_2) = Ax_2$

→ $Ax = b$ posses infinite many Sol

→ it is not one - one

↳

eg:-

$T: \mathbb{R}^2 \to \mathbb{R}^2$

$T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}$

$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 4 \\ 8 \end{pmatrix}$

$x_1$: $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}\begin{pmatrix} 4 \\ 0 \end{pmatrix} = \begin{pmatrix} 4 \\ 8 \end{pmatrix}$

$x_2$: $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}\begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 8 \end{pmatrix}$ → same output

$Ax = b$

if $Ax_1 = b$, $Ax_2 = b$ } then it is not one-one

$x + 2y = 4$

$2x + 4y = 8$

$\begin{pmatrix} 1 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$

$\begin{pmatrix} 1 & 2 & 4 \\ 0 & 0 & 0 \end{pmatrix}$

$x = 4 - 2y$

$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 4-2y \\ y \end{pmatrix}$

$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 4 \\ 0 \end{pmatrix} + y\begin{pmatrix} -2 \\ 1 \end{pmatrix}$

*2

## Onto

$T(x) = Ax$

$Ax = b$ ( $\forall b \rightarrow$ the system must be consistent )

$\Rightarrow$ Each row of A has pivot.

**Crux:-**

**Bijective:-**

one - one $\longrightarrow$ each col has pivot

onto $\rightarrow$ each row has pivot

$T : R^3 \rightarrow R^2 \rightarrow (2 \times 3) \rightarrow \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$

$T : R^2 \rightarrow R^3 \rightarrow (3 \times 2) \rightarrow \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix}$

$m > n \rightarrow$ each row can't have pivot $\rightarrow$ onto (not)

$m < n \rightarrow$ each col cant have pivot $\rightarrow$ 1-1 (not)

$m = n \rightarrow$ possibility

↳ inverse of square matrices exist

a transformation from $R^n \rightarrow R^m$, it is either both 1-1 & onto or else it cannot be

A matrix obtained from the identity matrix by performing single row operation is called elementary matrix

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

$R_2 + (-3)R_1$

$$B \begin{pmatrix} 1 & 2 \\ 0 & -2 \end{pmatrix} = B$$

$-\frac{1}{2} R_2$

$$B \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} = C$$

$R_1 - 2(R_2)$

$$R \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = D$$

$E_3 E_2 E_1 (D)$

$\downarrow$

$E_3 E_2 E_1 (I)$

$E_3 E_2 E_1 A = I$

$E_3 E_2 E A A^{-1} = I A^{-1}$

$E_3 E_2 E_1 (I) = A^{-1}$

$(A : I)$

$(I : A^{-1})$

## Recalling Lecture:-

→ inverse of bijective exist →

- 1-1 → pivot Cols
- onto → pivot Rows

$$\underset{2\times 3}{(\ )} \rightarrow \underset{3\times 2}{(\ )}$$

$$T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x+2y \\ 2x+7y \end{pmatrix}$$

$$A = \begin{pmatrix} 1 & 2 \\ 2 & 7 \end{pmatrix}$$

$$T \circ S(x) = I = S \circ T(x)$$

$$T \circ T^{-1}(x)$$

$$T\left(T^{-1}\begin{pmatrix} x \\ y \end{pmatrix}\right)$$

$$T\begin{pmatrix} \frac{7}{3}x - \frac{2}{3}y \\ -\frac{2}{3}x + \frac{1}{3}y \end{pmatrix}$$

$$T\begin{vmatrix} \frac{7}{3}x - \frac{2}{3}y - \frac{4}{3} + \frac{2}{3}y \\ \frac{14}{3}x - \frac{4}{3}y - \frac{14}{3}x + \frac{7}{3}y \end{vmatrix}$$

* finding inverse:-

$$\left(\begin{array}{cc:cc} 1 & 2 & 1 & 0 \\ 2 & 7 & 0 & 1 \end{array}\right)$$

$$\left(\begin{array}{cc:cc} 1 & 2 & 1 & 0 \\ 0 & 3 & -2 & 1 \end{array}\right)$$

$$\left(\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 1 & -\frac{2}{3} & \frac{1}{3} \end{array}\right)$$

$$\left(\begin{array}{cc|cc} 1 & 0 & \frac{7}{3} & -\frac{2}{3} \\ 0 & 1 & -\frac{2}{3} & \frac{1}{3} \end{array}\right)$$

* writing ans

$$T^{-1}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \frac{7}{3}x - \frac{2}{3}y \\ -\frac{2}{3}x + \frac{1}{3}y \end{pmatrix}$$

$$T\begin{bmatrix} x \\ y \end{bmatrix}$$

## Quiz Question

$a+bx = 1 = 1+0x$

$x = 0+1x$

$T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} a \\ a-b \end{pmatrix}$

① Basis of $P_1 = \{1, x\}$

② Basis of $R^2$ $\left\{\begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 3 \\ 4 \end{pmatrix}\right\}$

$T(1) \longrightarrow \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

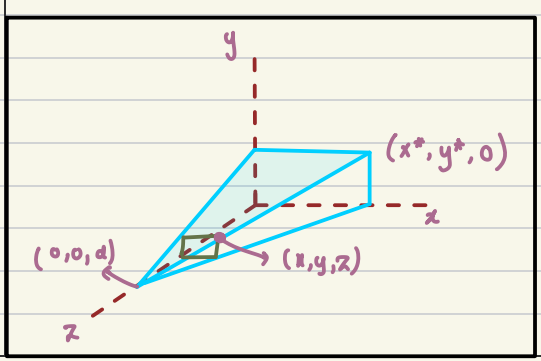
$T(x) \longrightarrow \begin{pmatrix} 0 \\ -1 \end{pmatrix}$

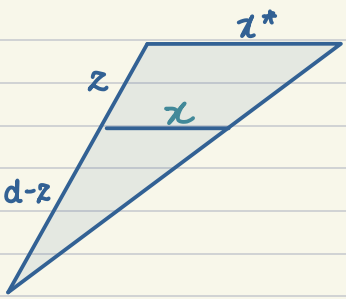
③ linear combinations + find coordinate vectors.

## Perspective Projections:-

→ 3D image on 2D

→ like a picture






by similar triangles

$$\frac{x^*}{d} = \frac{x}{d-z}$$

$$x^* = \frac{dx}{d-z}$$

$$x^* = \frac{x}{1-z/d}$$

$$y^* = \frac{y}{1-z/d}$$

$$P\begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -1/d & 1 \end{bmatrix} = \begin{pmatrix} x \\ y \\ 0 \\ 1-z/d \end{pmatrix}$$

## Determinants:-

$\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ : $ad - bc$ → area of parallelogram → $(2 \times 2)$

$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$ → volume parallelepiped → $(3 \times 3)$

why is $2\times2 \rightarrow$ -ve

is $2\times2 \rightarrow 0 \longrightarrow$ multiple line

$3\times3 \rightarrow$ -ve

$3\times3 \rightarrow 0 \longrightarrow$ vector lies on plane

$$\det(A) = -2\begin{vmatrix} 4 & 6 \\ 7 & 9 \end{vmatrix} + 5\begin{vmatrix} 1 & 3 \\ 7 & 9 \end{vmatrix} - 8\begin{vmatrix} 1 & 3 \\ 4 & 6 \end{vmatrix}$$

determinant of triangular

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 7 & 0 \\ 4 & 8 & 9 \end{bmatrix} \quad \begin{bmatrix} 1 & 4 & 5 \\ 0 & 7 & 3 \\ 0 & 0 & 9 \end{bmatrix} \longrightarrow$$ product of diagonal entries

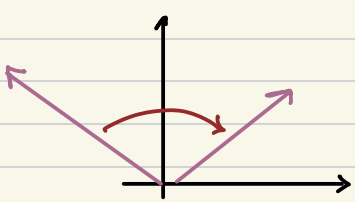
→ row interchange ⟶ (-minus sign)

→ row replacement (translate) (no effect)

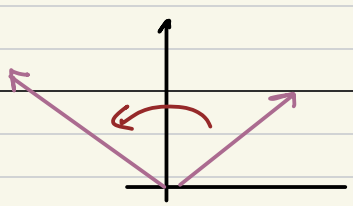
→ row scaling ⟶ (also divide outside)

why a determinant is negative.

$2\times2 \rightarrow$ Area of llgm



-ve (clockwise) $3\times3 \rightarrow$ volume of llgm



+ve (anticlockwise)

determinant algorithm :-

(multiplication)

$2 \times 2 \longrightarrow 2(2 \times 2) \rightarrow 2 \rightarrow 2!$

$3 \times 3 \rightarrow 3(2 \times 2) \rightarrow 6 \rightarrow 3!$

$4 \times 4 \rightarrow 4(3 \times 3) \rightarrow 24 \rightarrow 4!$

$5 \times 5 \rightarrow 5(4 \times 4) \rightarrow 120 \rightarrow 5!$

worst algo ever.

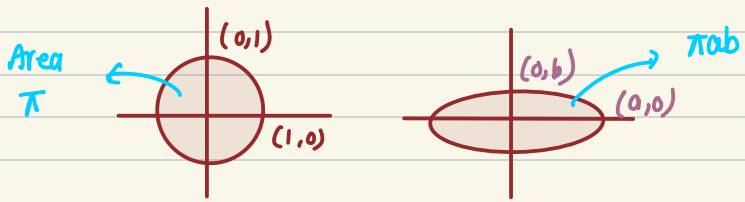
$100 \times 100 \longrightarrow 100!$

4) $\frac{100^{148}}{12 \times 30 \times 60 \times 60 \times 24} \longrightarrow 10^{141}$ years

comparison :-

| determinant $(n!)$ | | row operation $(n^3)$ |
|---|---|---|
| $2! \rightarrow 2$ | good for v-small | $2^3 \rightarrow 8$ |
| $3! \rightarrow 6$ | | $3^3 \rightarrow 24$ |
| $4! \rightarrow 24$ | | $4^3 \rightarrow 64$ |
| $5! \rightarrow 120$ | | $5^3 \rightarrow 125$ |
| $6! \rightarrow 720$ | | $6^3 \rightarrow 216$ |

Area of transformed figure = (Area of original figure)[|det A|]




$T\begin{pmatrix}0\\1\end{pmatrix} = \begin{pmatrix}0\\b\end{pmatrix} \longrightarrow \begin{bmatrix}a & 0\\0 & b\end{bmatrix}$

$T\begin{pmatrix}1\\0\end{pmatrix} = \begin{pmatrix}a\\0\end{pmatrix}$

Crammer's Rule, Inversion Method
→ determinant involed
→ not every matrix is invertible
→ works for squared systems.

# Chapter #5:-

## Eigen Values and Eigen Vectors

$T: \mathbb{R}^n \rightarrow \mathbb{R}^n$

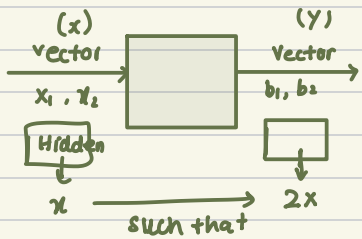
operator → when domain & co-domain are same

$T(x) = y$

$T(x_1) = a$

$T(x_2) = b$

$T(x_3) = I$




→ Eigen Vector
- → multiple of original vector

→ Eigen Values:-
- → magnitude of Eigen vector
- → ratio length of transform to length of original

reflection about
x-axis, y-axis
↓
If stays on axis on same side +1
if stay on axis on opposite side -1

→ eigen values are for square systems
→ the vector and eigen vector lie on the same line (parallel / non-parallel)
→ eigen value → length of transformed vector

## mathematical calculation of Eigen Vectors:-

$T: \mathbb{R}^n \longrightarrow \mathbb{R}^n$

$T(x) = Ax$

$T(x) = \lambda x$

$Ax = \lambda x$

$Ax - \lambda x = 0$

$x(A - \lambda) = 0$

$x(A - \lambda I) = 0$

$x \neq 0$ because $T(0) = 0$

→ homogenous System

↳ Sol of homogenous eq.

→ $x \Rightarrow$ eigen vector

* $T(0) = 0$

* $Bx = 0$

↓

non trivial Sol

↓

infinit many Sol

↓

↓

at least one non-pivot

↓

multiple

↓

not 1-1

↓

Det $B = 0$

$(A - \lambda I)x = 0$

For non-trivial

$\det(A - \lambda I) = 0$

$\lambda$ values ← $\lambda_1$, v ✓✓✓

↳ then $x =$ ✓✓✓

**Q#1:-**

$A\begin{pmatrix} 3 & -2 \\ 1 & 0 \end{pmatrix}, v = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$

Is v an eigen vector of A?

$T(v) = \lambda v$

$Av = \lambda v$

$$Av = \begin{bmatrix} 3 & -2 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

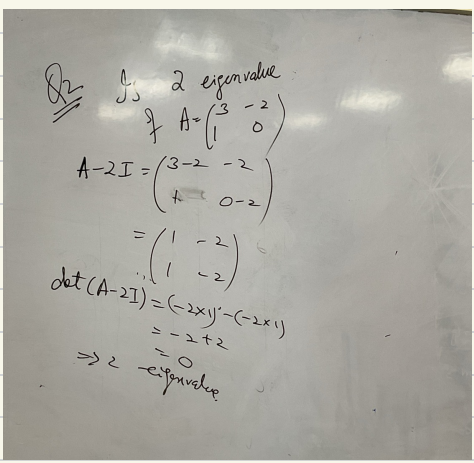
$$Av = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$$

$$A\begin{pmatrix} 2 \\ 1 \end{pmatrix} = 2\begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

→ v is eigenvector of A

**Q#2**

Is 2 eigen value of $A\begin{pmatrix} 3 & -2 \\ 1 & 0 \end{pmatrix}$



Q#3

Find all possible eigen values

$$A = \begin{pmatrix} 3 & -2 \\ 1 & 0 \end{pmatrix}$$

$$A - \lambda I = \begin{pmatrix} 3 & -2 \\ 1 & 0 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 3-\lambda & -2 \\ 1 & 0-\lambda \end{pmatrix}$$

$$\det(A - \lambda I) = (3-\lambda)(0-\lambda) - (-2 \times 1)$$

$$= -3\lambda + \lambda^2 + 2$$

$$\det(A - \lambda I) = 0$$

$$\lambda^2 - 3\lambda + 2 = 0$$

$$(\lambda - 2)(\lambda - 1) = 0$$

$$\lambda = 2 \quad \lambda = 1$$

Q#4 find all possible eigen vectors.

$\lambda = 1$

$$Ax = \lambda x$$

$$(A - \lambda I)x = 0$$

$$(A - I)x = 0$$

$$\begin{pmatrix} 2 & -2 & 0 \\ 1 & -1 & 0 \end{pmatrix}$$

echelon

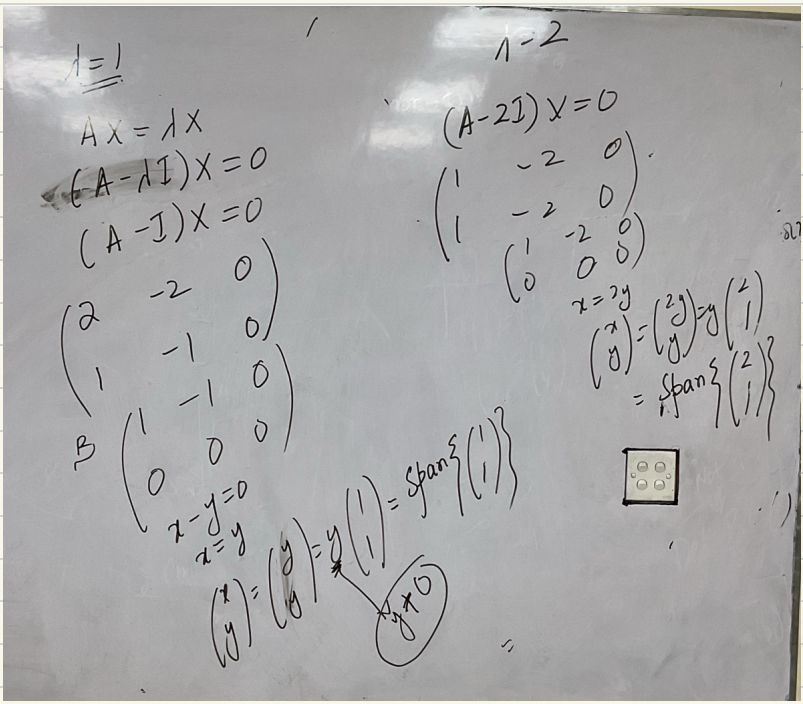
$$\begin{pmatrix} 1 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$x - y = 0$$

$$x = y$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} y \\ y \end{pmatrix} = y \left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right.$$

$$\text{span} \left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\}$$




(Online Lecture)

$$A = \begin{pmatrix} 0 & 6 & 8 \\ \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{2} & 0 \end{pmatrix} \qquad v = \begin{pmatrix} 16 \\ 4 \\ 1 \end{pmatrix}$$

$$Av = \begin{pmatrix} 0 & 6 & 8 \\ \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{2} & 0 \end{pmatrix} \begin{pmatrix} 16 \\ 4 \\ 1 \end{pmatrix} = \begin{pmatrix} 32 \\ 8 \\ 2 \end{pmatrix} = 2v$$

eigenvector of $A$, with eigenvalue $\lambda = 2$.

$$A = \begin{pmatrix} 2 & 2 \\ -4 & 8 \end{pmatrix} \qquad v = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$1\begin{pmatrix} 2 \\ -4 \end{pmatrix} + 1\begin{pmatrix} 2 \\ 8 \end{pmatrix}$$

$$\begin{pmatrix} 4 \\ 4 \end{pmatrix} = 4\begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

eigen value

+ve eigen value, both vectors lie

A $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ B $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$ C $\begin{pmatrix} -1 \\ 1 \end{pmatrix}$ D $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$ E $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$ ✗

no, null vector can't be

T $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

**Question:** Is $\lambda = 3$ an eigenvalue of $A = \begin{pmatrix} 2 & -4 \\ -1 & -1 \end{pmatrix}$?

In other words, does $Av = 3v$ have a nontrivial solution?
...does $Av - 3v = 0$ have a nontrivial solution?
...does $(A - 3I)v = 0$ have a nontrivial solution?
We know how to answer that! Row reduction!

$$A - 3I = \begin{pmatrix} 2 & -4 \\ -1 & -1 \end{pmatrix} - 3\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} -1 & -4 \\ -1 & -4 \end{pmatrix}$$

Row reduce:

$$\begin{pmatrix} -1 & -4 \\ -1 & -4 \end{pmatrix} \rightsquigarrow \begin{pmatrix} 1 & 4 \\ 0 & 0 \end{pmatrix}$$

Parametric form: $x = -4y$; parametric vector form: $\begin{pmatrix} x \\ y \end{pmatrix} = y\begin{pmatrix} -4 \\ 1 \end{pmatrix}$.

Does there exist an eigenvector with eigenvalue $\lambda = 3$? Yes! Any nonzero multiple of $\begin{pmatrix} -4 \\ 1 \end{pmatrix}$. Check:

$$\begin{pmatrix} 2 & -4 \\ -1 & -1 \end{pmatrix}\begin{pmatrix} -4 \\ 1 \end{pmatrix} = \begin{pmatrix} -12 \\ 3 \end{pmatrix} = 3\begin{pmatrix} -4 \\ 1 \end{pmatrix}. \checkmark$$

→ 0 vector cannot be an eigen vector,
→ therefore, we are interesented in non-trivial solution

Eigen Spaces:-
→ we are solving a homogenous system, therefore its solution is a nullspace
↳ key idea → is a space.

→ find all possible $\lambda_i$

set of all eigen vectors also has 0 vector, but we don't include 0,

→ to make it a space.

Nullspace (Sol of eigen vector) ∪ (0)

Find a basis for the 2-eigenspace of

$$A = \begin{pmatrix} 4 & -1 & 6 \\ 2 & 1 & 6 \\ 2 & -1 & 8 \end{pmatrix}.$$

$$A - 2I = \begin{pmatrix} 2 & -1 & 6 \\ 2 & -1 & 6 \\ 2 & -1 & 6 \end{pmatrix} \xrightarrow{\text{row reduce}} \begin{pmatrix} 1 & -\frac{1}{2} & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\xrightarrow{\text{parametric form}} x = \frac{1}{2}y - 3z$$

$$\xrightarrow{\text{parametric vector form}} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = y \begin{pmatrix} \frac{1}{2} \\ 1 \\ 0 \end{pmatrix} + z \begin{pmatrix} -3 \\ 0 \\ 1 \end{pmatrix}$$

$$\xrightarrow{\text{basis}} \left\{ \begin{pmatrix} \frac{1}{2} \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -3 \\ 0 \\ 1 \end{pmatrix} \right\}$$

(transform vector) (simple vector)

$Av \rightarrow v$

→ lies on same plane

Let $A$ be an $n \times n$ matrix and let $\lambda$ be a number.

1. $\lambda$ is an eigenvalue of $A$ if and only if $(A - \lambda I)x = 0$ has a nontrivial solution, if and only if $\text{Nul}(A - \lambda I) \neq \{0\}$.
2. In this case, finding a basis for the $\lambda$-eigenspace of $A$ means finding a basis for $\text{Nul}(A - \lambda I)$ as usual, i.e. by finding the parametric vector form for the general solution to $(A - \lambda I)x = 0$.
3. The eigenvectors with eigenvalue $\lambda$ are the nonzero elements of $\text{Nul}(A - \lambda I)$, i.e. the nontrivial solutions to $(A - \lambda I)x = 0$.

**Fact:** The eigenvalues of a triangular matrix are the diagonal entries.

**Why?** $\text{Nul}(A - \lambda I) \neq \{0\}$ if and only if $A - \lambda I$ is not invertible, if and only if $\det(A - \lambda I) = 0$.

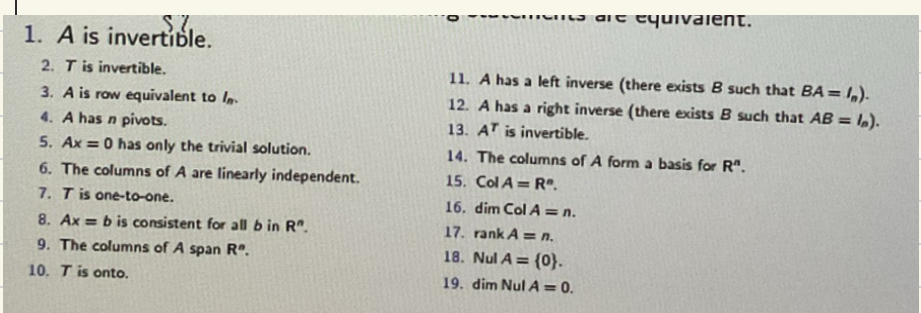
$$\begin{pmatrix} 3 & 4 & 1 & 2 \\ 0 & -1 & -2 & 7 \\ 0 & 0 & 8 & 12 \\ 0 & 0 & 0 & -3 \end{pmatrix} - \lambda I_4 = \begin{pmatrix} 3-\lambda & 4 & 1 & 2 \\ 0 & -1-\lambda & -2 & 7 \\ 0 & 0 & 8-\lambda & 12 \\ 0 & 0 & 0 & -3-\lambda \end{pmatrix}.$$

The determinant is $(3 - \lambda)(-1 - \lambda)(8 - \lambda)(-3 - \lambda)$, which is zero exactly when $\lambda = 3, -1, 8$, or $-3$.

if a matrix is invertible, Eigen value $\neq 0$
if a matrix is not invertible, eign value $= 0$

→ distinct eigen values.

→ linearly independent
→ do not lie on span of eachother



1. $A$ is invertible.
2. $T$ is invertible.
3. $A$ is row equivalent to $I_n$.
4. $A$ has $n$ pivots.
5. $Ax = 0$ has only the trivial solution.
6. The columns of $A$ are linearly independent.
7. $T$ is one-to-one.
8. $Ax = b$ is consistent for all $b$ in $\mathbb{R}^n$.
9. The columns of $A$ span $\mathbb{R}^n$.
10. $T$ is onto.
11. $A$ has a left inverse (there exists $B$ such that $BA = I_n$).
12. $A$ has a right inverse (there exists $B$ such that $AB = I_n$).
13. $A^T$ is invertible.
14. The columns of $A$ form a basis for $\mathbb{R}^n$.
15. $\text{Col}\, A = \mathbb{R}^n$.
16. $\dim \text{Col}\, A = n$.
17. $\text{rank}\, A = n$.
18. $\text{Nul}\, A = \{0\}$.
19. $\dim \text{Nul}\, A = 0$.

determinant $\neq 0$
eigen value $\neq 0$

finding $\lambda$ :-

$$\det(A - \lambda I) = 0$$

Trace = Sum of eigen values
Determinant = Product of eigen values.

**Definition**
The **trace** of a square matrix $A$ is $\mathrm{Tr}(A)$ = sum of the diagonal entries of $A$.

**Shortcut**
The characteristic polynomial of a $2 \times 2$ matrix $A$ is
$f(\lambda) = \lambda^2 - \mathrm{Tr}(A)\,\lambda + \det(A)$.

if 2x2 ⟶ polynomial of degree 2
n – eigen values

Algebraic Multiplicity:-
→ how many time a root appear.

→ distant Eigen value
⟶ Algebraic multiplicity = 1

**Poll**
If you count the eigenvalues of $A$, with their algebraic multiplicities, you will get:
A. Always $n$.
B. Always at most $n$, but sometimes less.
C. Always at least $n$, but sometimes more.
D. None of the above.

## Diagonalization:-

→ calculating higher power of matrix.

→ power of diagonal matrix is power of diagonal entriess.

$$D^n = \begin{bmatrix} 2^n & 0 \\ 0 & 4^n \end{bmatrix}$$

## Similarity:-

$$A = PBP^{-1}$$

(137) → slide

### Powers of Matrices that are Similar to Diagonal Ones

What if $A$ is not diagonal?

Example

Let $A = \begin{pmatrix} 1 & 2 \\ -1 & 4 \end{pmatrix}$. Compute $A^n$.

$$A = PDP^{-1} \quad \text{where} \quad P = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \quad \text{and} \quad D = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}.$$

Then

$$A^2 = (PDP^{-1})(PDP^{-1}) = PD(P^{-1}P)DP^{-1} = PDIDP^{-1} = PD^2P^{-1}$$

$$A^3 = (PDP^{-1})(PD^2P^{-1}) = PD(P^{-1}P)D^2P^{-1} = PDID^2P^{-1} = PD^3P^{-1}$$

$$\vdots$$

$$A^n = PD^nP^{-1}$$

D → matrix of eigen values
P → matrix of eigen vectors
} same order

diagonalization

→ n x n matrix
→ n linearly independent Eigen vectors.

## Diagonalization

Example

**Problem:** Diagonalize $A = \begin{pmatrix} 1 & 2 \\ -1 & 4 \end{pmatrix}$.

The characteristic polynomial is

$$f(\lambda) = \lambda^2 - \mathrm{Tr}(A)\,\lambda + \det(A) = \lambda^2 - 5\lambda + 6 = (\lambda - 2)(\lambda - 3).$$

Therefore the eigenvalues are 2 and 3. Let's compute some eigenvectors:

$$(A - 2I)x = 0 \iff \begin{pmatrix} -1 & 2 \\ -1 & 2 \end{pmatrix} x = 0 \overset{\text{rref}}{\rightsquigarrow} \begin{pmatrix} 1 & -2 \\ 0 & 0 \end{pmatrix} x = 0$$

The parametric form is $x = 2y$, so $v_1 = \binom{2}{1}$ is an eigenvector with eigenvalue 2.

$$(A - 3I)x = 0 \iff \begin{pmatrix} -2 & 2 \\ -1 & 1 \end{pmatrix} x = 0 \overset{\text{rref}}{\rightsquigarrow} \begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix} x = 0$$

The parametric form is $x = y$, so $v_2 = \binom{1}{1}$ is an eigenvector with eigenvalue 3.

The eigenvectors $v_1, v_2$ are linearly independent, so the Diagonalization Theorem says

$$A = PDP^{-1} \quad \text{for} \quad P = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \qquad D = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}.$$

## Diagonalization

Another example, continued

**Problem:** Diagonalize $A = \begin{pmatrix} 4 & -3 & 0 \\ 2 & -1 & 0 \\ 1 & -1 & 1 \end{pmatrix}$.

Now let's compute the 2-eigenspace:

$$(A - 2I)x = 0 \iff \begin{pmatrix} 2 & -3 & 0 \\ 2 & -3 & 0 \\ 1 & -1 & -1 \end{pmatrix} x = 0 \overset{\text{rref}}{\rightsquigarrow} \begin{pmatrix} 1 & 0 & -3 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{pmatrix} x = 0$$

The parametric form is $x = 3z, y = 2z$, so an eigenvector with eigenvalue 2 is

$$v_3 = \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}.$$

The eigenvectors $v_1, v_2, v_3$ are linearly independent: $v_1, v_2$ form a basis for the 1-eigenspace, and $v_3$ is not contained in the 1-eigenspace. Therefore the Diagonalization Theorem says

$$A = PDP^{-1} \quad \text{for} \quad P = \begin{pmatrix} 1 & 0 & 3 \\ 1 & 0 & 2 \\ 0 & 1 & 1 \end{pmatrix} \qquad D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}.$$

**Note:** In this case, there are three linearly independent eigenvectors, but only two distinct eigenvalues.

## Diagonalization

Procedure

**How to diagonalize a matrix $A$:**

1. Find the eigenvalues of $A$ using the characteristic polynomial.
2. For each eigenvalue $\lambda$ of $A$, compute a basis $\mathcal{B}_\lambda$ for the $\lambda$-eigenspace.
3. If there are fewer than $n$ total vectors in the union of all of the eigenspace bases $\mathcal{B}_\lambda$, then the matrix is not diagonalizable.
4. Otherwise, the $n$ vectors $v_1, v_2, \ldots, v_n$ in your eigenspace bases are linearly independent, and $A = PDP^{-1}$ for

$$P = \begin{pmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & & | \end{pmatrix} \quad \text{and} \quad D = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix},$$

where $\lambda_i$ is the eigenvalue for $v_i$.

# (Online Session-2)

recall:-

① $(A-\lambda I) = 0$

② $|A-\lambda I|$ ⟶ characteristic equation (in terms of $\lambda$)

degree of Poly = size of matrix

③ put characteristic equation $= 0$ ⟶ value of $\lambda$

fundamental rule of Algebra — number of roots = polynomial degree

⟶ multiplicity:-

$P_n$ = n-roots

- repeated roots
- distinct roots (multiplicity = 1)

multiplicity → number of times a root is repeated.

(algebraic) ↳ term associated with eigan values

geometric multiplicity:-

→ dimension of eiganspace

→ related to eigan vectors

number of vectors in eigen space → geometric multiplicity

↓

dimension of eigen space

diagonalization (eigan decompositim)

→ (to factorize a matrix)

① eigan values

② eigan space

③ n total vector → n eigan values

④

$$P = (v_1, v_2, v_3) \qquad D = \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{pmatrix}$$

$$A = PDP^{-1}$$

Uses:-

* power of matrix $A^3 \rightarrow \begin{bmatrix} (\lambda_1)^3 & 0 \\ 0 & (\lambda_2)^3 \end{bmatrix}$

* Simplify linear transformation

* differential equations.

curse of dimensionality (>_<)

## Principal Composition Analysis:- (PCA)

project data higher → lower dimension

we project in into subspace of eigan vectors

→ it preserves

- → data matrix
- → co ? matrix
- → subspace
- → project onto it.

## Linear discriminant Analysis

→ eigan values

### Eigenvalues and Eigenvectors of $A^n$

If $\lambda$ is the eigenvalue and $v$ be the corresponding eigenvector of the matrix $A$, then

$$Av = \lambda v$$

We want to find eigenvalue and eigenvector of $A^2$

$$A^2v = A(Av) = A(\lambda v) = \lambda Av = \lambda(\lambda v) = \lambda^2 v$$

$\lambda^2$ is eigenvalue of $A^2$ and $v$ is the corresponding eigenvector

$$A^n v = \lambda^n v$$

# Linear Recurrence Relation :-

Sequence of numbers

recurrence of order k :-
requires k initial conditions

**Linear Recurrence Relation**

Let $(x_n) = (x_0, x_1, x_2, \dots)$ be a sequence of numbers that is defined as follows:

1. $x_0 = a_0, x_1 = a_1, \dots, x_{k-1} = a_{k-1}$, where $a_0, a_1, \dots, a_{k-1}$ are scalars.
2. For all $n \geq k, x_n = c_1 x_{n-1} + c_2 x_{n-2} + \cdots + c_k x_{n-k}$ where $c_1, c_2, \dots, c_k$ are scalars.

If $c_k \neq 0$, the equation in (2) is called a linear recurrence relation of order $k$. The equations in (1) are referred to as the initial conditions of the recurrence.

**Examples**

- order 2 — $x_{n+2} = x_{n+1} + x_n$, $x_0 = 1, x_1 = 1$. ⟶ initial condition
- order 1 — $x_{n+1} = 2x_n$, $x_0 = 3$.

linear recurrence in matrix form :-

$$(x_{n+2}) = a x_{n+1} + b x_n \quad , \quad x_1 = c_1, \; x_0 = c_0$$

add Slide

Ginonacci ⟶ each term is avg of pre 2 terms

① matrix
② eigan vector & value
③ diagonalize

a : trace
c : determinant.

$$\lambda^2 + a\lambda + b = 0$$

$$\lambda_1 \neq \lambda_2$$

**Important Result**

**Theorem**

Let $x_n = ax_{n-1} + bx_{n-2}$ be a recurrence relation. Let $\lambda_1$ and $\lambda_2$ be the eigenvalues of the associated characteristic equation $\lambda^2 - a\lambda - b = 0$.

1. If $\lambda_1 \neq \lambda_2$, then
$$x_n = c_1\lambda_1^n + c_2\lambda_2^n$$
for some scalars $c_1$ and $c_2$.
2. If $\lambda_1 = \lambda_2$, then
$$x_n = (c_1 + c_2 n)\lambda^n$$
for some scalars $c_1$ and $c_2$.

## Examples

Solve the following recurrence relation with the given initial conditions.

$$y_1 = 1, y_2 = 6, y_k = 4y_{k-1} - 4y_{k-2}, \quad k \geq 3.$$

Characteristics equation $\lambda^2 - 4\lambda + 4 = 0$.
Solution of the quadratic equation is Eigenvalues: $\lambda_1 = 2, \lambda_1 = 2$. So,

$$y_k = c_1(2)^k + c_2 k 2^k.$$

because of repeated roots

As, $y_1 = 1$, so, $2c_1 + 2c_2 = 1$, $y_2 = 6$, so, $4c_2 + 8c_2 = 6$.
Solution of above system is

$$c_1 = -\frac{1}{2}, \quad c_2 = 1.$$

Hence,

$$y_k = -\frac{1}{2}2^k + k2^k$$

if a third repeated then $(+c_3 k^2 2^k)$

## Practice Problems

1. Solve the recurrence relation with the given initial conditions.
   - $a_0 = 4, a_1 = 1, a_n = a_{n-1} - \frac{a_{n-2}}{4}$, for $n \geq 2$.
   - $a_n = 2a_{n-1} + a_{n-2} - 2a_{n-3}$, subject to $a_0 = 2, a_1 = 2, a_2 = 4$, for $n \geq 3$.
2. Find the limiting values of $y_k$ and $z_k$, $(k \to \infty)$ if

$$\begin{aligned} y_{k+1} &= .8y_k + .3z_k, \quad y_0 = 0 \\ z_{k+1} &= .2y_k + .7z_k, \quad z_0 = 5. \end{aligned}$$

$A^n x_0 = P D^n P^{-1} x_0$

$A_{2\times 2} \Rightarrow$ 2 linearly indep eigan values / vectors $\Big\}$ basis of $\mathbb{R}^2$

diagonalizable matrix gives us basis formed by eigan vectors

$A^n x_0 \qquad x_0 = c_1 v_1 + c_2 v_2$

$A^2 x_0 = c_1 A^2 v_1 + c_2 A^2 v_2$

$= c_1 (\lambda_1^2 v_1) + c_2 (\lambda_2^2 v_2)$

$A^n x_0 = c_1 \lambda_1^n v_1 + c_2 \lambda_2^n v_2$

audio-based slide

## Lecture Review:-

* domain = co-domain
* $\det(A - \lambda I) = 0$ → characteristic eq
  degree ↗ == size of matrix.
* $A - \lambda I : 0$ → vectors
* 1 col never has pivot. ??

$$A \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix} \longrightarrow \text{linearly dependents.}$$
$$\det(A) = 0$$

→ $\lambda = 0$

→ $\sum \lambda_i$ = trace (A)
↳ sum of diagonal matrix.

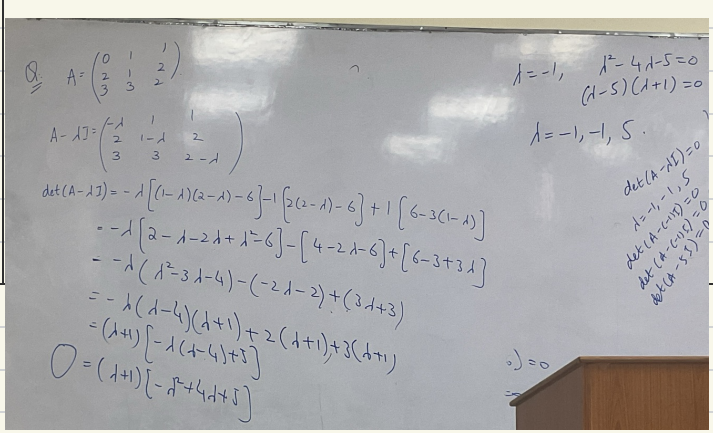
$\pi \lambda_i$ = det (A)

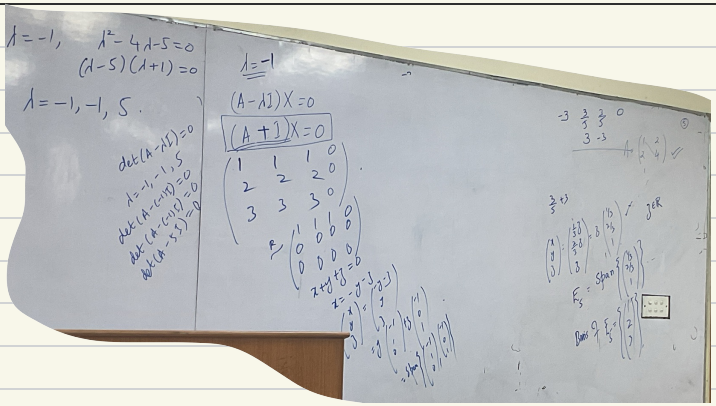
characteristic eq:-

trace (A) $\lambda^2$ + det $\lambda$ (A) +

$$(A - \lambda I) X = 0$$

→ alway 1 non pivot





# Eigen Decomposition / Diagonalization :-

$$A = PDP^{-1}$$

## Linear recurrence and diagnolization.

method 1 :-

$$x_n = a_1 x_{n-1} + a_2 x_{n-2} + \dots + a_k x_{n-k}$$

$$x_n = 2x_{n-1} + 3x_{n-2} + 4x_{n-3}$$

$$x_{n-1} = x_{n-1}$$

$$x_{n-2} = x_{n-2}$$

$$\begin{pmatrix} x_n \\ x_{n-1} \\ x_{n-2} \end{pmatrix} = \begin{pmatrix} 2 & 3 & 4 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} x_{n-1} \\ x_{n-2} \\ x_{n-3} \end{pmatrix}$$

$$X_n = A X_{n-1}$$

$$X_2 = \begin{pmatrix} x_2 \\ x_1 \\ x_0 \end{pmatrix} = \begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix}$$

$X_n = A X_{n-1}$

$F_n = F_{n-1} + F_{n-2}$

$F_{n+1}$

$X_3 = AX_2$

$X_4 = AX_3 = A^2 X_2$

$X_n = A^{n-2} X_2$

$$\boxed{X_n = PD^{n-2}P^{-1}X_2}$$ → Marcov chain

method 2 :-

$\lambda^3 - 2\lambda^2 - 3\lambda - 4 = 0$

$\lambda = \checkmark, \checkmark, \checkmark$

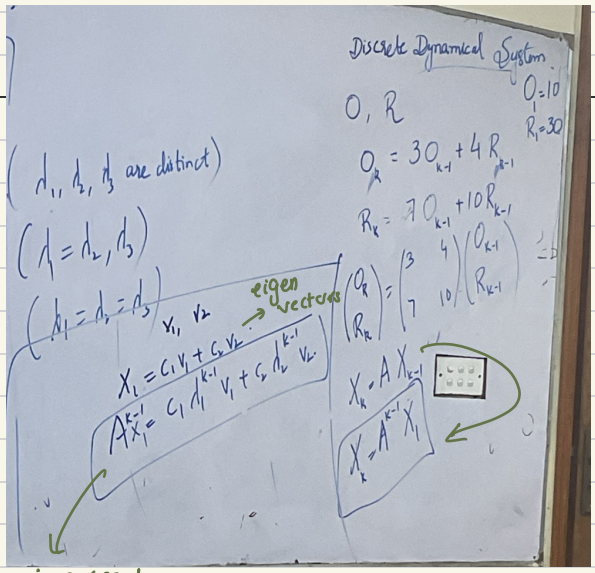
$x_n = c_1\lambda_1^n + c_2\lambda_2^n + c_3\lambda_3^n$ $(\lambda_1, \lambda_2, \lambda_3)$ → distinct.

$x_n = (c_1 + c_2 n)\lambda_1^n + c_3\lambda_2^n$ $(\lambda_1 = \lambda_2, \lambda_3)$

$x_n = (c_1 + c_2 n + c_3 n^2)\lambda_1^n$ $(\lambda_1 = \lambda_2 = \lambda_3)$

difference equation:-

→ discrete dyanamical system




method 2:-

→ eigen vector make basis of vectors

$$A^{n-1} X_1$$

$$X_1 = c_1 v_1 + c_2 v_2$$

$$A^{n-1} X_1 = c_1 A^{n-1} v_1 + c_2 A^{n-1} v_2$$

$$X_n = c_1 \lambda_1^{n-1} v_1 + c_2 \lambda_2^{n-1} v_2$$

$(\lambda_1, \lambda_2)$ → distinct

Question 1:-

$$f_n = f_{n-1} + f_{n-2}, \quad f_0 = 1, \quad f_1 = 2$$

$$f_n - f_{n-1} - f_{n-2} = 0$$

$$\lambda^2 - \lambda - 1 = 0$$

$$\lambda = \frac{1 \pm \sqrt{1+4}}{2}$$

$$\lambda_1 = \frac{1+\sqrt{5}}{2}, \quad \lambda_2 = \frac{1-\sqrt{5}}{2}$$

$$f_n = c_1 \lambda_1^n + c_2 \lambda_2^n$$

$$f_n = c_1 \left(\frac{1+\sqrt{5}}{2}\right)^n + c_2 \left(\frac{1-\sqrt{5}}{2}\right)^n$$

$$f_0 = c_1 \left(\frac{1+\sqrt{5}}{2}\right)^0 + c_2 \left(\frac{1-\sqrt{5}}{2}\right)^0$$

$$1 = c_1 + c_2$$

$$1 = c_1 \left(\frac{1+\sqrt{5}}{2}\right) + c_2 \left(\frac{1-\sqrt{5}}{2}\right)$$

$$c_1 \left(\frac{1+\sqrt{5}}{2}\right) + \left(\frac{1+\sqrt{5}}{2}\right) c_2 = \left(\frac{1+\sqrt{5}}{2}\right)$$

$$c_1 \left(\frac{1+\sqrt{5}}{2}\right) \ominus \left(\frac{1-\sqrt{5}}{2}\right) c_2 = \oplus 1$$

$$\sqrt{5}\, c_2 = -\frac{1+\sqrt{5}}{2} \implies c_2 = \frac{5-\sqrt{5}}{10}$$

$$c_1 = 1 - c_2 = 1 - \left(\frac{5-\sqrt{5}}{10}\right) = \Rightarrow c_1 = \frac{5+\sqrt{5}}{10}$$

**Question 2:**

$x_n = 2x_{n-1} - x_{n-2}$ $\quad x_0 = 0$, $x_1 = 1$

$x_n - 2x_{n-1} + x_{n-2} = 0$

$\lambda^2 - 2\lambda + 1 = 0$

$\lambda^2 - \lambda - \lambda + 1$

$\lambda(\lambda-1) - 1(\lambda-1) = 0$

(1) find eigen value put exp = 0

$(\lambda-1)(\lambda-1) = 0$

$\lambda = 1$

(2) make eq

$x_n = (c_1 + c_2 n)\lambda^n$

(3) put equal to given values

$x_n = c_1 \lambda^n + c_2 n \lambda^n$

$0 = c_1 (1)^0 + c_2 (0)(1)^0$

$0 = c_1$

$1 = c_1 (1)^1 + c_2 (1)(1)^1$

$1 = c_1 + c_2$

$1 = c_2$

(4) Solve linear system

$c_1 = 0$

$c_2 = 1$

(5) find $x_n$ by placing values

$x_n = (0 + 1(n))(1)^n$

$x_n = n$

**Recurrance**

$\lambda_1, \lambda_2, \lambda_3$

$x_n = c_1 \lambda_1^n + c_2 \lambda_2^n + c_3 \lambda_3^n$

$\lambda_1, \lambda_2, \lambda_2$

$x_n = c_1 \lambda_1^n + c_2 \lambda_2^n + c_3 n \lambda_2^n$

$\lambda_1, \lambda_2, \lambda_3$

$x_n = c_1 \lambda_1^n + c_2 n \lambda_1^n + c_3 n^2 \lambda_1^n$

**Proof:-**

$$\begin{pmatrix} x^{n+1} \\ x_n \end{pmatrix} = c_1 \lambda_1^{n+1} v_1 + c_2 \lambda_2^{n+1} v_2$$

$$\begin{pmatrix} x^{n+1} \\ x_n \end{pmatrix} = c_1 \lambda_1^{n+1} \begin{pmatrix} 1 \\ ? \end{pmatrix} + c_2 \lambda_2^{n+1} \begin{pmatrix} 1 \\ ? \end{pmatrix}$$

$$\begin{pmatrix} x^{n+1} \\ x_n \end{pmatrix} = \begin{pmatrix} c_1 \lambda_1^{n+1} + c_2 \lambda_2^{n+1} \\ c_1 \lambda_1^{n+1} ? + c_2 \lambda_2^{n+1} ? \end{pmatrix}$$

$$x^{n+1} = c_1 \lambda_1^{n+1} + c_2 \lambda_2^{n+1}$$
$$x_n = c_1 \lambda_1^{n} + c_2 \lambda_2^{n}$$

proof of equating MI & MII

**Proof:-**

Jordan form:-

$A = PDP^{-1}$ (if not diagonalizable)

$A = PJP^{-1}$

$A_{3\times 3}$

$\lambda_1$ $\quad \lambda_2, \lambda_1$

$v_1$ $\quad v_2$

$$P\begin{pmatrix} \downarrow & \downarrow & ? \\ v_1 & v_2 & \\ 1 & 1 & \end{pmatrix}$$

$(A - \lambda_2 I)v_2 = v_2$

Solve

$$J\begin{pmatrix} \lambda_1 & 1 & 0 \\ 0 & \lambda_2 & 1 \\ 0 & 0 & \lambda_3 \end{pmatrix}$$

## Difference Equations:-

$$x_n = a x_{n-1} + b y_{n-1}$$

$$y_n = c x_{n-1} + d y_{n-1}$$

$$X_n = A^{n-1} X_1$$

$$X_n = c_1 \lambda^{n-1} v_1 + c_2 \lambda^{n-1} v_2$$

| | | $n \to \infty$ |
|---|---|---|
| attractor | $0 \leq \lambda \leq 1$ | Sol $\to \infty$ |
| repeller | $\lambda > 1$ | Sol $\to 0$ |
| saddle | $\lambda = 1$ | Sol $\to$ constant |

?? → increase in 1 value
→ decrease in other value.

## System of differential equations:-
(continuous)

$$\frac{dy}{dt} = 3y \qquad T(y) = 3y$$

$$\int \frac{1}{y}\, dy = \int 3\, dt$$

$$\ln(y) = 3t + c_1$$

$$y = e^{3t + c_1}$$

$$y = e^{3t} \cdot e^{c_1}$$

$$y = c_2 e^{3t} \longrightarrow$$

system:-

$$\frac{dx}{dt} = ax + by$$

$$\frac{dy}{dt} = cx + dy.$$

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

$$X' = AX$$

Put $X = PS$

$$(PS)' = APS$$
$$PS' = APS$$
$$S' = P^{-1}APS$$
$$S' = DS$$

RW

$$X' = AX$$
$$X' = PDP^{-1}X$$
$$P^{-1}X' = DP^{-1}X$$
$$(P^{-1}X)' = D(P^{-1}X)$$

$$P^{-1}X = S$$
$$S' = DS$$

$$\begin{pmatrix} u' \\ v' \end{pmatrix} = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix}$$

$$u' = \lambda_1 u$$
$$v' = \lambda_2 v$$

$u' = \lambda_1 u$
$u = c_1 e^{\lambda_1 t}$

$v' = \lambda_2 v$
$v = c_2 e^{\lambda_2 t}$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ v_1 & v_2 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = c_1 e^{\lambda_1 t} v_1 + c_2 e^{\lambda_2 t} v_2$$

**discrete system**

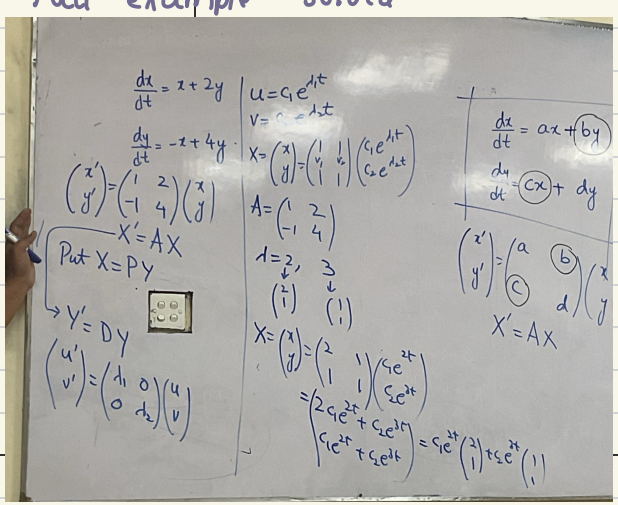
$$x_{n+1} = ax_n + by_n$$
$$y_{n+1} = cx_n + dy_n$$

$$x_n = c_1 \lambda_1^n v_1 + c_2 \lambda_2^n v_2$$

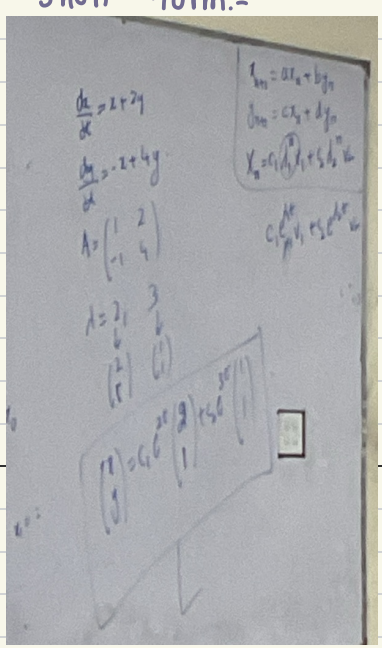
**continuous system**

$$x_n = c_1 e^{\lambda_1 t} v_1 + c_1 e^{\lambda_2 t} v_2$$

full example solved




short form:-



# Chapter #6:-

## inner product:-

$$\vec{u} = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}, \quad \vec{v} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$$

$$\vec{u} \cdot \vec{v} = u_1 v_1 + u_2 v_2$$

$$= \begin{pmatrix} v_1 & + v_2 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$$

$$u = \begin{pmatrix} u_1 \\ \vdots \\ u_n \end{pmatrix}, \quad v = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix}$$

$$u \cdot v = \sum_{i=1}^{n} u_i v_i$$

$$u \cdot v = v^T u \longrightarrow \begin{pmatrix} v_1 & v_2 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$$

## Length:-

$$\vec{u} \cdot \vec{u} = |\vec{u}|^2$$

$$|\vec{u}| = \sqrt{\vec{u} \cdot \vec{u}}$$

→ in higher dimension → norm $\|u\|$

# Inner Product

* **Dot Product**

$$\vec{u} = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}, \quad \vec{v} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$$

$$\vec{u} \cdot \vec{v} = u_1 v_1 + u_2 v_2$$

→ Angle
⇒ Distance

1. $$u = \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{pmatrix}, \quad v = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix}$$

$$u \cdot v = u_1 v_1 + u_2 v_2 + \cdots u_n v_n$$

$$u \cdot v = \sum_{i=1}^{n} u_i v_i$$

2. **length / Norm** → in higher dimension

$$\|u\| = \sqrt{u \cdot u}$$

→ Co-relation:-
→ Co-variance

3. **perpendicular / Orthogonal:-**

$$u \cdot v = 0$$

4. **angle:**

$$\cos\theta = \frac{\vec{u}\cdot\vec{v}}{|\vec{u}||\vec{v}|}$$

5. **orthogonal set:-**

$$\{v_1, v_2 \ldots v_n\}$$

$$v_i \cdot v_j = 0 \quad , \forall\, i \neq j$$

(Set of mutually perpendicular vectors)

**Theorem:-** An orthogonal set of non-zero vectors is linearly independent

$$\{v_1, v_2 \ldots v_n\}$$

$$c_1v_1 + c_2v_2 + \ldots\ldots c_nv_n = 0$$

$$c_1 v_1 v_1 + c_2 v_1 v_2 \ldots\ldots c_n v_1 v_n = 0$$

($v_1 v_2 = 0$, $v_1 v_n = 0$)

$$c_1 \|v\|^2 = 0$$

$$c_1 = 0 \qquad v_1 \neq 0 \quad \|v\| \neq 0$$

if $c_1$, $c_2$ and $c_3$ is zero

$\Rightarrow$ def, if all the scalars are zero, then vectors are linearly independent.

Stack exchange

Steve ⟶ for DS & ML

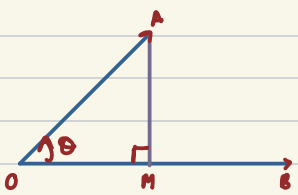
6. **Orthonormal Set:-**

$$v_i \cdot v_j = \begin{cases} 0 & i \neq j \\ 1 & i = j \end{cases}$$

(Set of mutually perpendicular unit vectors)

**Projection:-**

Projection of $\vec{a}$ along $\vec{b}$ $= \left(\frac{\vec{a}\cdot\vec{b}}{|\vec{b}|}\right)\hat{b}$




$$\frac{|OM|}{|OA|} = \cos\theta$$

$$|OM| = a\cos\theta$$

$$|OM| = \frac{|a||b|\cos\theta}{|b|}$$

$$= \frac{a\cdot b}{|b|}$$

$$= \frac{\vec{a}\cdot\vec{b}}{|\vec{b}|}\cdot\frac{\vec{b}}{|\vec{b}|}$$

$$= \frac{\vec{a}\cdot\vec{b}}{|\vec{b}|^2}$$

$$= \frac{\vec{a}\cdot\vec{b}}{\vec{b}\cdot\vec{b}}\cdot\vec{b}$$

## Projection matrix:-

$$\frac{a \cdot b}{b \cdot b} \; b$$

$$= \frac{b^T a}{b^T b} \cdot b$$

$$= \frac{b\,(b^T a)}{b^T b}$$

$$= \frac{(b\,b^T)}{b^T b}\, a$$

$$P = \left(\frac{b\,b^T}{b^T b}\right)$$

↳ projection matrix

$b_{n\times 1} \quad b^T_{1\times n} = \underbrace{n \times n}_{\text{matrix}}$

$b^T_{1\times n} \quad b_{n\times 1} = \underbrace{1\times 1}$

↳ scalar

↳ neutralizing factor

**Rank of matrix**

→ (always 1)

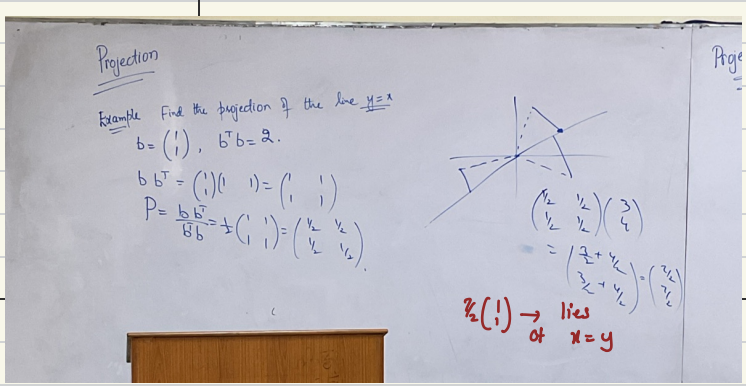
→ because it is projected on line

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} \begin{pmatrix} x & y & z \end{pmatrix}$$

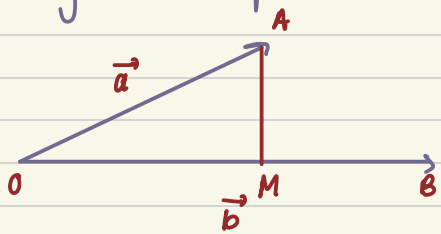
$$\begin{pmatrix} xx & xy & xz \\ yx & yy & yz \\ zx & zy & zz \end{pmatrix}$$

↳ multiples

exam



## Orthogonal Decomposition:-




$\vec{OM} + \vec{OA} = \vec{a}$

$\vec{OM} = \vec{a} - \text{proj}_{\vec{b}}\ \vec{a}$

$F = F_x \hat{i} + F_y \hat{j}$

$F \cdot i = F_x$

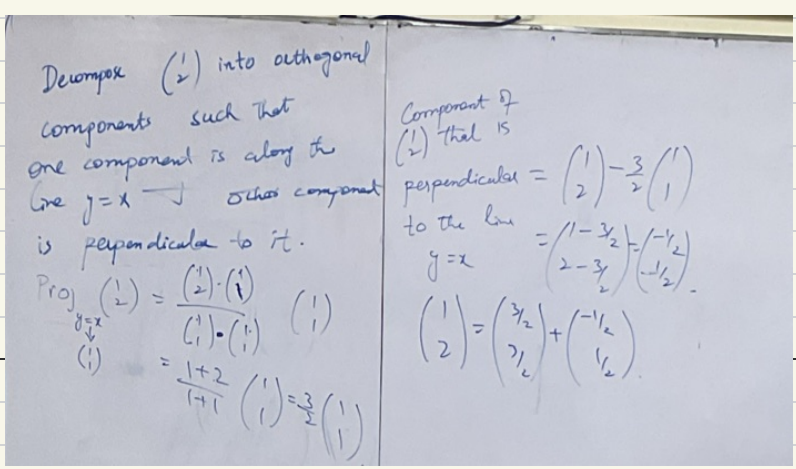
$\frac{\vec{a} \cdot \vec{b}}{\vec{b} \cdot \vec{b}}\ \vec{b}$

$\vec{b}$ is unit vector

$(\vec{a} \cdot \vec{b}) \cdot \vec{b}$

component of $\vec{a}$ along $\vec{b}$.

Decompose $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ into orthogonal components such that one component is along the line $y = x$ & other component is perpendicular to it.

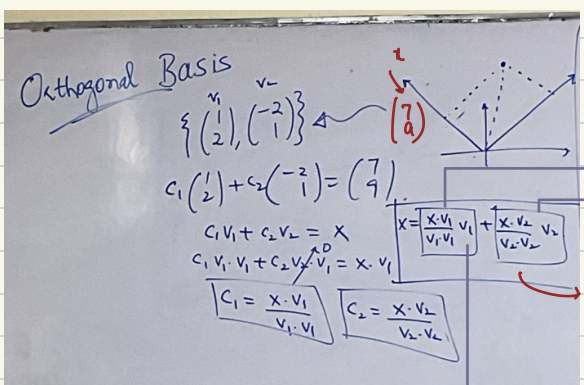


# Orthogonal Basis:-

→ if basis are orthogonal

$$\left\{ \begin{pmatrix} 1 \\ 2 \end{pmatrix}_{a} \cdot \begin{pmatrix} -2 \\ 1 \end{pmatrix}_{b} \right\} \longrightarrow \begin{pmatrix} 7 \\ 9 \end{pmatrix}_{x}$$

$a \cdot b = 0$

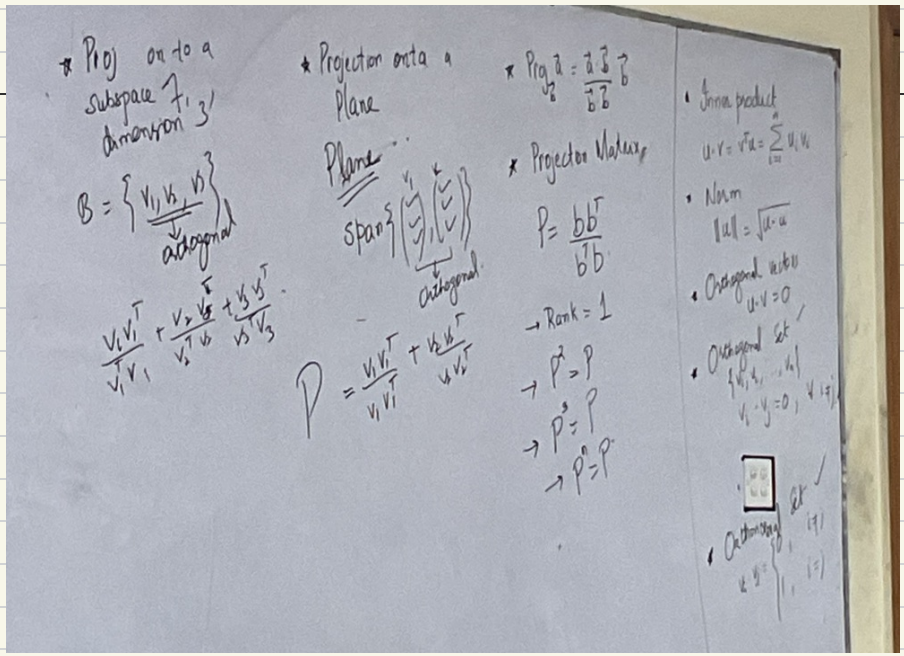



→ proj x along $v_1$

→ proj x along $v_2$

→ geometric interpretation of linear combination of vectors.

projection matrix




making vectors orthogonal

$x - 2y - z = 0$

$z = x - 2y$

$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{Bmatrix} x \\ y \\ x-2y \end{Bmatrix}$

$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = x\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + y\begin{pmatrix} 0 \\ 1 \\ -2 \end{pmatrix}$

solve for this.

let's make it orthogonal

Q → finding a projection matrix, that projects onto the plane $x-2y-z=0$

(can make any one of them)

$$v_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \qquad v_2 = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$v_2 \cdot v_1 = 0$$

$$x + z = 0$$
$$x = -z$$
$$x - 2y - z$$

$$\begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & -2 & -1 & 0 \end{bmatrix}$$

$R_2 - R_1$

$$\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & -2 & -2 & 0 \end{bmatrix}$$

$$-2y = 2z$$
$$y = -z$$
$$x = -z$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -z \\ -z \\ z \end{pmatrix} = z \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix}$$

Basis: $\left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix} \right\}$

→ put this vector into plane eq.

→ for orthogal check dot prod = 0

$$\frac{v_1 v_1^T}{v_1^T v_1} + \frac{v_2 v_2^T}{v_2^T v_2}$$

$$\frac{\begin{pmatrix}1\\0\\1\end{pmatrix}\begin{pmatrix}1 & 0 & 1\end{pmatrix}}{\begin{pmatrix}1 & 0 & 1\end{pmatrix}\begin{pmatrix}1\\0\\1\end{pmatrix}} + \frac{\begin{pmatrix}-1\\-1\\1\end{pmatrix}\begin{pmatrix}-1 & -1 & 1\end{pmatrix}}{\begin{pmatrix}-1 & -1 & 1\end{pmatrix}\begin{pmatrix}-1\\-1\\1\end{pmatrix}}$$

$$\frac{1}{2}\begin{pmatrix}1 & 0 & 1\\0 & 0 & 0\\1 & 0 & 1\end{pmatrix} + \frac{1}{3}\begin{pmatrix}1 & 1 & -1\\1 & 1 & -1\\-1 & -1 & 1\end{pmatrix}$$

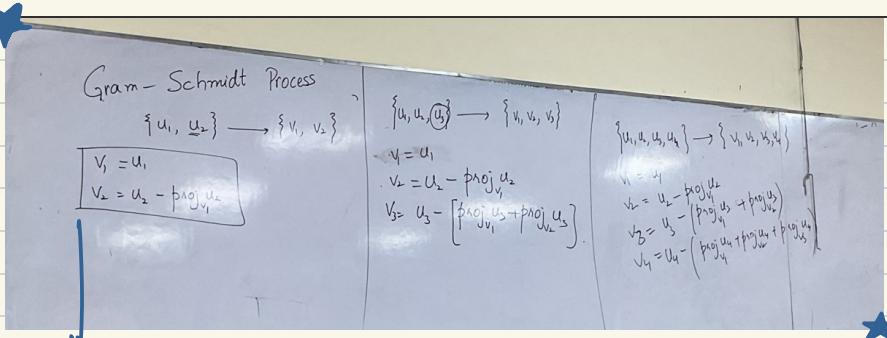
→ every basis cannot be orthogonal so we study an algo called gram shmith algorith to convert any basis into an orthogonal one.

## Gram - Schmidt Process:

$$\{u_1, u_2\} \longrightarrow \{v_1, v_2\}$$

$$v_1 = u_1$$

$$v_2 = u_2 - \text{proj}_{v_1} u_2$$




→ these are orthogonal

↳ to convert them into orthonormal

↓

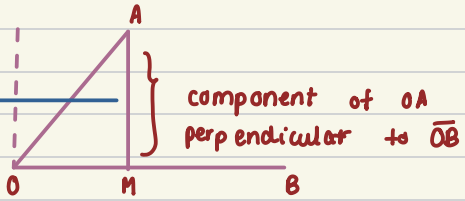
divide by magnitude of norm

$\|v_1\|$

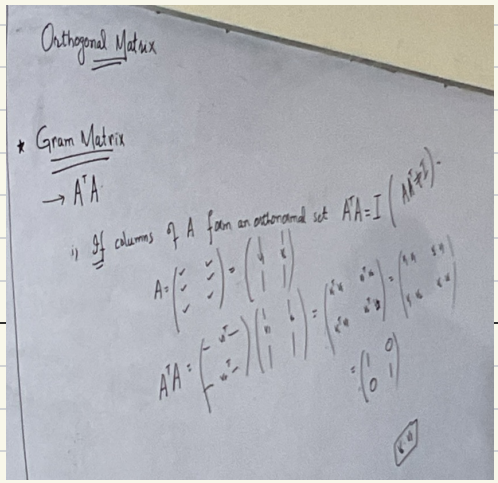
find point that is closest to the plane.

↙

(projection)




Orthogonal:-



## Gram Matrix:-

→ $A^TA \rightarrow I$

↳ enteries of gram matrix → (represent projection)

If a is a square matrix and (orthonormal)

$$A^TA = I = AA^T$$

↳ orthogonal matrix (represented by Q)

properties:-

$$Q^TQ = QQ^T = I$$

$$Q^{-1} = Q^T$$

det $\pm 1$

+1 → rotation

-1 → Reflection

$$Q^TQ = I$$

$$\det(Q^TQ) = \det(I)$$

$$\det(Q^T)\det(Q) = \det(I)$$

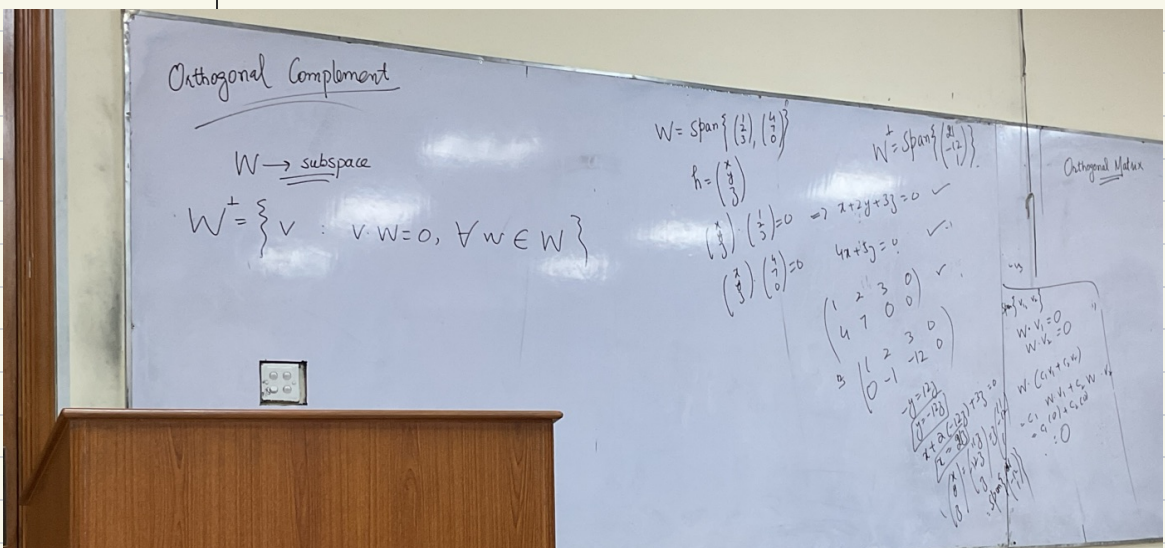
$$\det(Q)^2 = 1$$

$$\det Q = \sqrt{1}$$

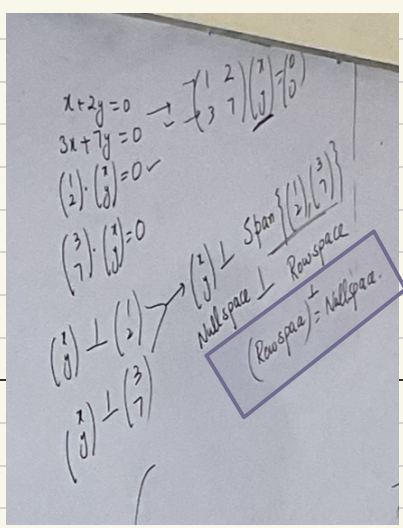
$$\det Q = \pm 1$$

hi muneeba how are you ok bye armish is awesome!
im fine thank you and you
ASAP pinut butter yammy

## Orthogonal Complement :-




verification:-



Recommended :-

$W = \text{span}\left\{\begin{pmatrix}1\\2\\3\end{pmatrix}\begin{pmatrix}4\\7\\0\end{pmatrix}\right\}$

$W^{\perp} = ?$

$W = \text{RowSpace}(A)$

$A = \begin{pmatrix}1 & 2 & 3\\ 4 & 7 & 0\end{pmatrix}$

$Ax = 0$

$\hookrightarrow \begin{pmatrix}x\\y\\z\end{pmatrix}.$

1. if basis are given, check if they are orthogonal or not.
2. make orthogonal
3. find projection matrix & multiply with vector (to be projected)

# Least Squares

⇒ Regression :-

→ $Ax = b$
→ $Ax = \lambda x$
→ to approximate the sol if inconsistent system $Ax = b$

$Ax = b$
↓
$\hat{x}$ = Proj b Col(A)

$Ax = b$ is consistent, iff $b$ lies in CSpace $(A)$.

$$A = \left\{\begin{pmatrix}1\\2\end{pmatrix}, \begin{pmatrix}2\\4\end{pmatrix}\right\} \quad b = \begin{bmatrix}9\\2\end{bmatrix}$$

$y = 2x$

$Ax = b$

$$\begin{pmatrix}1 & 2 & 9\\2 & 4 & 2\end{pmatrix} \approx \begin{pmatrix}1 & 2 & 9\\0 & 0 & -16\end{pmatrix}$$ → inconsistent.

$Ax = b$

$$\begin{pmatrix}1 & 2\\2 & 4\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix} = \begin{pmatrix}9\\2\end{pmatrix}$$ replacing b

$$\text{proj}_{\text{Col A}}\, b = \text{proj}_{\begin{bmatrix}1\\2\end{bmatrix}} \begin{bmatrix}9\\2\end{bmatrix}$$

$$= \frac{\begin{pmatrix}9\\2\end{pmatrix}\begin{pmatrix}1\\2\end{pmatrix}}{\begin{pmatrix}1\\2\end{pmatrix}\begin{pmatrix}1\\2\end{pmatrix}} \begin{pmatrix}1\\2\end{pmatrix} = \begin{pmatrix}13/5\\26/5\end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}\begin{pmatrix} \hat{x} \\ \hat{y} \end{pmatrix} = \begin{pmatrix} 13/5 \\ 26/5 \end{pmatrix}$$

$$\left(\begin{array}{cc|c} 1 & 2 & 13/5 \\ 2 & 4 & 26/5 \end{array}\right) \sim \left(\begin{array}{cc|c} 1 & 2 & 13/5 \\ 0 & 0 & 0 \end{array}\right)$$

$$\hat{x} + 2\hat{y} = 13/5$$

$$\hat{x} = \frac{13}{5} - 2\hat{y}$$

$$\begin{pmatrix} \hat{x} \\ \hat{y} \end{pmatrix} = \begin{pmatrix} \frac{13}{5} - 2\hat{y} \\ \hat{y} \end{pmatrix}$$

$$= \begin{pmatrix} 13/5 \\ 0 \end{pmatrix} + \hat{y}\begin{pmatrix} -2 \\ 1 \end{pmatrix}$$

$\Rightarrow$ Logical steps:

$Ax = b$ (inconsistent)

$A\hat{x} = \text{Proj}_{\text{col}(A)}\, b$ — (A)

$\text{Proj}_{\text{col}(A)}\, b = \text{col}(A)$

$b - \text{proj}_{\text{col}(A)}\, b \in [\text{Col}(A)]^{\perp}$

$[\text{col}(A)]^{\perp} = \text{NullSpace}(A^T)$

$\Rightarrow b - \text{Proj}_{\text{col}(A)}\, b \in \text{NullSpace}(A^T)$

if $\begin{pmatrix} 1 \\ 2 \end{pmatrix} \in \text{NullSpace}(A^T)$

$Ax = 0$

$A^T\begin{pmatrix} 1 \\ 2 \end{pmatrix} = 0$

so we can write as:-

$$A^T\left(b - \underset{col(A)}{proj}\, b\right) = 0$$

from (A)

$$A^T(b - A\hat{x}) = 0$$

$$A^Tb - A^TA\hat{x} = 0$$

$$(A^TA)\hat{x} = A^Tb$$

⇒ Crux:- $Ax = b$

$(A^TA)\hat{x} = A^Tb$

$(A^TA)\hat{x} = A^Tb$

$A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$, $A^T \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$

$A^TA = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$

$= \begin{pmatrix} 5 & 10 \\ 10 & 20 \end{pmatrix}$

$A^Tb = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}\begin{pmatrix} 9 \\ 2 \end{pmatrix} = \begin{pmatrix} 13 \\ 26 \end{pmatrix}$

putting in eqs:-

$$(A^TA)\hat{x} = A^Tb$$

$$\begin{pmatrix} 5 & 10 \\ 10 & 20 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 13 \\ 26 \end{pmatrix}$$

# Least Squares

$$Ax = b$$

$$(A^T A)\hat{x} = A^T b \quad - (B)$$

* $A_{m \times n}$ , $A^T_{n \times m}$

* $A^T A$ ($n \times m$ $m \times n$) $\rightarrow n \times n$

  ↳ Gram matrix

* $(A^T A)^T = A^T (A^T)^T \rightarrow A^T A$

  ↳ symmetric

if $A^T A$ is invertible

from b:- $\hat{x} = (A^T A)\, A^T b$

$(A^T A)^{-1} = A^{-1} (A^T)^{-1}$ ⟶ if rectangular

↳ incorrect

then:-

$(AB)^{-1} = B^{-1} (A)$ ⟶ if they are square matrix

therefore we calculate

left Inverse $A^{-1}_{\text{left}} = (A^T A)^{-1} A^T$

why is it called left inverse?

$\rightarrow A^{-1}_{left} A$

$\rightarrow (A^TA)^{-1}A^TA$

$\rightarrow (A^TA)^{-1}(A^TA)$

$\rightarrow (A)^{-1}(A)$

$\rightarrow I \quad \rightarrow (A)^{-1}_{Left} A = I$

* $A^TA$ is invertible iff Col(A) are linearly independent.

* $A = \begin{pmatrix} 1 & 3 & 5 \\ 2 & 4 & 9 \end{pmatrix}$

if $A_{m \times n}$ if $n > m$

Col A linearly dependent.

* $A_{m \times n}$ if $n \leq m \longrightarrow$ **Full Column Rank**

**Crux 1:-**

Left inverse of A exist iff A has left column rank

(Col(A) are L. I)

$A^{-1}_{left} = (A^TA)^{-1}A^T$

$A^{-1}A = I$

**Crux 2:-**

Right inverse of exist iff A has full row rank

(Row (A) are L.I

$A^{-1}_{right} = A^T(AA^T)^{-1}$

$A\,A^{-1}_{right} = I$

$A\hat{x} = b$

↓ put value

$A(A^TA)^{-1}A^Tb$

↓

$\hat{x} \longrightarrow$

$A \cdot A^{-1}_{left} \Rightarrow$ Projection

Projection Matrix $= A(A^TA)A^T$

multiplying it by b

↳ projects it onto A

## Side-by-Side Analysis

$W = \left\{\begin{pmatrix}1\\2\end{pmatrix}, \begin{pmatrix}3\\8\end{pmatrix}\right\}$

↳ GS → orthogonal

$P = \frac{v_1v_1^T}{v_1^Tv_1} + \frac{v_2v_2^T}{v_2^Tv_2}$

$P = A(A^TA)^{-1}A^T$

↙

no need to make orthogonal

(lecture)

Least Square:-

Projection ⟶

Using $AA^{-1}_{left} = P_{on\ column\ Space}$.

Using $A^{-1}_{right} A = P_{on\ Row\ Space}$.

→ if W has dependent columns

$W = Span\left\{\begin{pmatrix}1\\2\\3\\4\end{pmatrix}\begin{pmatrix}5\\6\\7\\8\end{pmatrix}\begin{pmatrix}4\\8\\10\\12\end{pmatrix}\right\}$ ↙ skip this

~~Let~~

Least Square:-

↳ $Ax = b$ (1) ⟶ by this

$(A^TA)\hat{x} = A^Tb$ (2) ⟶ by inverse method.

## Simple Linear Regression:-

find the best fit line passing throught the points (1,2), (3,4), (5,7) (8,9).

$2 = a+b$
$4 = a+3b$
$7 = a+5b$
$9 = a+8b$

$$\begin{bmatrix}1&1\\1&3\\1&5\\1&8\end{bmatrix}\begin{bmatrix}a\\b\end{bmatrix} = \begin{bmatrix}2\\4\\7\\9\end{bmatrix}$$

↓ response of system

find the best fit line

(1,7,9) (4,5,8) (2,9,7) (4,2,10) (9,2,4) (194,7)

$y = a + bx_1 + cx_1$

$$A = \begin{pmatrix}1&1&7\\1&4&5\\1&2&9\\1&4&2\\1&9&2\\1&10&4\end{pmatrix}\begin{pmatrix}a\\b\\c\end{pmatrix} = \begin{pmatrix}a\\8\\7\\16\\4\\7\end{pmatrix}$$

$y = A(A^TA)^{-1}A^Tb$

$A^T_{3\times6}\ A_{6\times3} = A^TA_{(3\times3)}$

$(A^TA)^{-1} \Rightarrow (A^TA)^{-1}_{(3\times3)}$

$(A^TA)^{-1}A^T = (A^TA)^{-1}A^T \times b$
$3\times3\quad 3\times6 \qquad 3\times6 \qquad (6\times1)$

equal to weights ⟵ Ans $(3\times1)$ ⟶ $y = ? + ?x_1 + ?? x_2$

Supervised learning ⟶ labelled data.

Unsupervised learning ⟶ unlabelled data.

Prediction ⟶ Regression

Classification ⟶ Linear Discrete Analysis.

Model ⟶ function.

linear function

$y = a+bx_1$ (line) in $R^2$

$y = a+bx_1 + cx_2$ plane

$y = a+bx_1 + cx_2 + dx_3$ hyper plan

$= (A^TA)^{-1}A^T \times b$

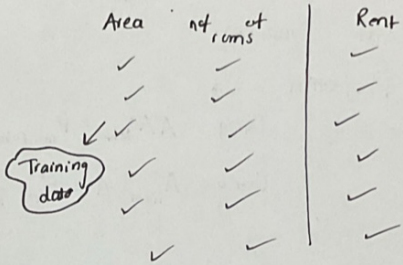
## Method 1:-

$(x_1, y_1), (x_2, y_2), (x_3, y_3)$

$y = a + bx$

$(y_1 - (a+bx_1))^2$
$+$
$(y_2 - (a+bx_2))^2 = F$
$+$
$(y_3 - (a+bx_3))^2$

Such that it is a minimum.




$y = \checkmark + \checkmark(\text{A}) + \checkmark\text{A}(\text{R})$

↳ predicts.

↳ $\frac{\partial F}{\partial a} = 0, \frac{\partial F}{\partial b} = 0 \longrightarrow (a, b) \rightarrow$ critical point

↳ gives normal equations.

↳ finding minimum

↳ hessian matrix (~~minima~~) → place here.

---

## Regression.

a line that has least distance from every data point.

$$\begin{pmatrix} 1 & x_1 \\ 1 & x_2 \\ 1 & x_3 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} \approx \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix}$$

features. inconsistent.

$A$ $x$ $b$

A → bias

b or c → weights

gives weights to features.

Vandermonde matrix

↳ numpy has its function.

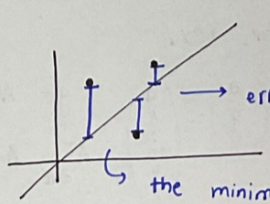
$\hat{x} = (A^T A)^{-1} A^T b$

↳ Linear Regression Line:-

$y = A(A^T A)^{-1} A^T b$

↳ projection matrix

↳ has least distance from line.



Why left inverse always exist?

→ over-determined. (tall)

$$\begin{bmatrix} 1 & 5 \\ 2 & 6 \\ 3 & 7 \\ 4 & 8 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix}$$

→ 4 eqs, 2 variables
↳ over-determined system.

chances ⇒ → left-inverse exists

→ under-determined (wide)

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$$

→ 2 eqs, 4 variable
↳ under-determined system

chances of right inverse

**Quadratic Regression:-**

$y = a + bx + cx^2$ → 1 parabola.

$$A = \begin{pmatrix} | & | & | \\ 1 & x_1 & x_1^2 \\ | & | & | \end{pmatrix}$$

# 2 parabolas

$$A = \begin{pmatrix} | & | & | & | & | \\ 1 & x_1 & x_1^2 & x_2 & x_2^2 \\ | & | & | & | & | \end{pmatrix}$$

**Linear Regression**

$y = a + bx$ $\quad y = (1\ x)\begin{pmatrix} a \\ b \end{pmatrix} \rightarrow y = \tilde{D}W$ (actual value)

$y = a + bx_1 + cx_2$ $\quad y = (1\ x_1\ x_2)\begin{pmatrix} a \\ b \\ c \end{pmatrix} \rightarrow y = \tilde{D}W$

$y = a + bx_1 + cx_2 + dx_3$ $\quad y = (1\ x_1\ x_2\ x_3)\begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} \rightarrow y = \tilde{D}W$

predicted
We used ~~modelled~~ value in ML models

$$\hat{Y} = \tilde{D}\left[(\tilde{D}^T\tilde{D})^{-1}\tilde{D}^T Y\right]$$

↓ hat matrix

(1,2)
(4,5)
(7,9)
(2,9)

Hat matrix
→ $H^2 = H$
→ eigen values 0 and 1
→ always symmetric

$A\hat{x} = b$

$\hat{x} = (A^TA)^{-1}A^Tb$

$\hat{y} = (\tilde{D}^T\tilde{D})^{-1}\tilde{D}^TY$

$Ax = b$

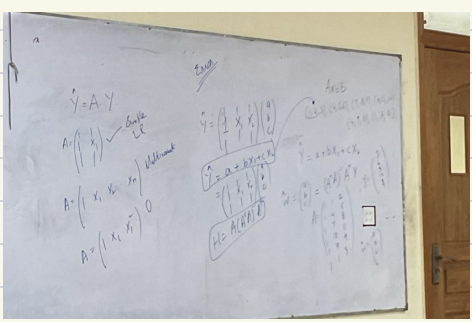
$(1,2,3), (2,5,6), (8,7,9)$ ⟶ best fit plane

$\hat{Y} = a + bx_1 + cx_2$

⇒ we do not do quadratic regression, because higher degree polynomials cause more fluctuations and cause issues in credibility of data. (over fitting problem)

⇒ Hasty

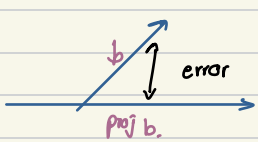
⇒



$b - \text{Proj}_A\, b$

$b - A(A^TA)^{-1}A^Tb$

hat matrix




$= b - Hb$

$= (I - H)b$

$= \|(I - H)b\|$ ⟶ length of the error.

(Multicolinearity):-

→ If matrix is ~~invertible~~ non-invertible, then features are dependent on eachother

↓

2 Solutions:-

① → skip the dependent column.

② → sometimes, vectors are closely related.

↳ used Ridge Regression.

$$\hat{W} = A\left(A^TA + \alpha I\right)^{-1}A^Tb$$

$\alpha > 0$

generalized inverse.

① → make independent
(controls multicollinearity)

② → shrinkage method.

QR - factorization:-

(reduces computational cost)

$A = QR$ → upper triangular matrix.

↓

orthogonal matrix. → columns are orthonormal

↓

dot of vectors = 0

itself dot = 0

$$A = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}$$

$v_1 = u_1$

$v_2 = u_2 - \text{proj}_{v_1} u_2$

$\longrightarrow \frac{v_1}{\|v_1\|}$

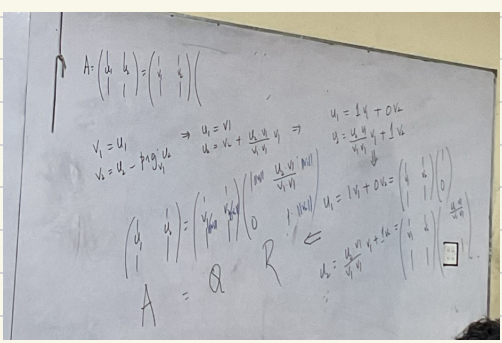
$$Q = \begin{pmatrix} | & | \\ \frac{v_1}{\|v_1\|} & \frac{v_2}{\|v_2\|} \\ | & | \end{pmatrix}$$

$A = QR \qquad Q^TQ = I$

$Q^TA = Q^TQR$

$R = Q^TA$

$$A = \begin{pmatrix} | & | \\ u_1 & u_2 \\ | & | \end{pmatrix} = \begin{pmatrix} | & | \\ v_1 & v_2 \\ | & | \end{pmatrix}$$



$v_1 = u_1$

$v_2 = u_2 - \text{proj}_{v_1} u_2$

$u_1 = v_1$

$u_2 = v_2 + \frac{v_1 \cdot u_2}{v_1 \cdot v_1}$ $(v_1)$

$u_1 = 1v_1 + 0v_2 \longrightarrow \begin{pmatrix} | & | \\ v_1 & v_2 \\ | & | \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix}$

$u_2 = \left(\frac{v_1 \cdot u_2}{v_1 \cdot v_1}\right) v_1 + 1v_2 \longrightarrow \begin{pmatrix} | & | \\ v_1 & v_2 \\ | & | \end{pmatrix}\begin{pmatrix} \frac{v_1 \cdot u_2}{v_1 \cdot v_1} \\ 1 \end{pmatrix}$

$$\begin{pmatrix} | & | \\ u_1 & u_2 \\ | & | \end{pmatrix} = \begin{pmatrix} | & | \\ v_1 & v_2 \\ | & | \end{pmatrix}\begin{pmatrix} 1 & \frac{v_1 \cdot u_2}{v_1 \cdot v_1} \\ 0 & 1 \end{pmatrix}$$

$Q$ orthogonal

$$\begin{pmatrix} | & | \\ u_1 & u_2 \\ | & | \end{pmatrix} = \begin{pmatrix} | & | \\ \frac{v_1}{\|v_1\|} & \frac{v_2}{\|v_2\|} \\ | & | \end{pmatrix} \begin{pmatrix} 1 \times \|v_1\| & \frac{v_1 \cdot u_2}{v_1 \cdot v_1} \|v_1\| \\ 0 & 1 \times \|v_2\| \end{pmatrix}$$

$$P = A\left(A^TA\right)^{-1}A^T$$

$$= (QR)\left((QR)^T(QR)\right)^{-1}(QR)^T$$

$$= (QR)\left(R^TQ^TQR\right)^{-1}(QR)^T$$

$$= (QR)(R^TR)^{-1}(QR)^T$$

$$= QRR^{-1}(R^T)^{-1}R^TQ^T$$

$$\boxed{P = QQ^T}$$ → projection matrix

$$\hat{y} = A(A^TA)^{-1}A^Tb$$

$$\boxed{\hat{y} = (QQ^T)b}$$

## Inner Product:-

calculates the distance between points

Inner Product:

$$(c_1u + c_2v)\cdot w = c_1u\cdot w + c_2u\cdot w$$

$$u\cdot v = v\cdot u$$

$$u\cdot u \geq 0 \qquad u\cdot u = 0 \iff u = 0$$

a function that take 2 vectors inputs and gives a real number that satisfies 3 properties

$$f: V \times V \longrightarrow \mathbb{R}$$

- $f(c_1u + c_2v, w) = c_1 f(u,w) + c_2 f(v,w)$
- $f(u,v) = f(v,u)$
- $f(u,u) \geq 0 \qquad f(u,u) = 0 \iff u = 0$

we use angular brackets to show that this function is inner product.

$$\langle \, , \, \rangle : V \times V \longrightarrow \mathbb{R}$$

- $\langle c_1u + c_2u_1, w \rangle = c_1 \langle u,w \rangle + c_2 \langle v, w \rangle$
- $\langle u,v \rangle = \langle u,v \rangle$
- $\langle u,u \rangle \geq 0$ && $\langle u,u \rangle = 0 \iff u = 0$

**Inner Product :-**

1. $\langle u,v \rangle = \sum_{i=1}^{n} u_i v_i$ (simple inner product)

$= v^T u$

check Rec - 1

2. $\langle u, v \rangle_W = v^T W u$ (Weight Inner Product)

$\rightarrow A^T A\hat{x} = A^T b$

$w \rightarrow$ weights

3. $\langle u, v \rangle_\theta = \cos\theta$

↳ cosine similarity

4. $\langle u, v \rangle_{\log} = \sum_{i=1}^{n} \log(1 + u_i v_i)$

↳ entropy calculation.

5. $\langle u, v \rangle_E = \sum_i \frac{u_i v_i}{1 + u_i^2 + v_i^2}$

↳ activation function

↳ main component activation → in neural networks

6. $\langle u, v \rangle_\Sigma = u^T \Sigma^{-1} v$

↳ used in clustering algorithm

↳ used in standard deviation.

7. $\langle u, v \rangle_K = k(u, v)$

$k(u,v) = (u^T v + c)^d$ ↳ kernel inner product.

$k(u,v) = e^{-\gamma \|u - v\|^2}$

8. $\langle u, v \rangle = \sum_{k=1}^{n} u[k] * v[k]$

↳ convolution inner product.

**Norm:-**

$$\|u\| = \sqrt{\langle u, u \rangle}$$

**(2-norm):**

* $\|u\|_2 = \sqrt{u_1^2 + u_2^2 + \dots u_n^2}$

**(1- norm):** → in lasso regression

* $\|u\|_1 = |u_1| + |u_2| + \dots |u_n|$

$\|u\|_p = \left( |u_1|^p + |u_2|^p + \dots |u_n|^p \right)^{1/p}$

↳ place 1, 2 becomes 1 and 2 norm

**(infinity norm):**

* $\|u\|_\infty = \max \{ |u_1|, |u_2|, \dots |u_n| \}$

* $\langle f, g \rangle = \int_a^b f(t)\, g(t)\, dt$ (continuous)

$\text{norm} = \sqrt{\int_a^b [f(t)]^2 dt}$

↳ for discrete fns →

$f \Rightarrow [0, 2]$

$g \Rightarrow [0, 2]$

$f \cdot g = f(1)g(1) + f(0)\, g(0) + f(2)g(2)$

but what if fns are continuous

$\sum_{i=0}^{2} f(i)\, g(i)$

for continuous s

$$\int_0^2 f(t)\, g(t)\, dt$$

$\langle A, B \rangle = \text{trace}(B^T A)$ (matrices)

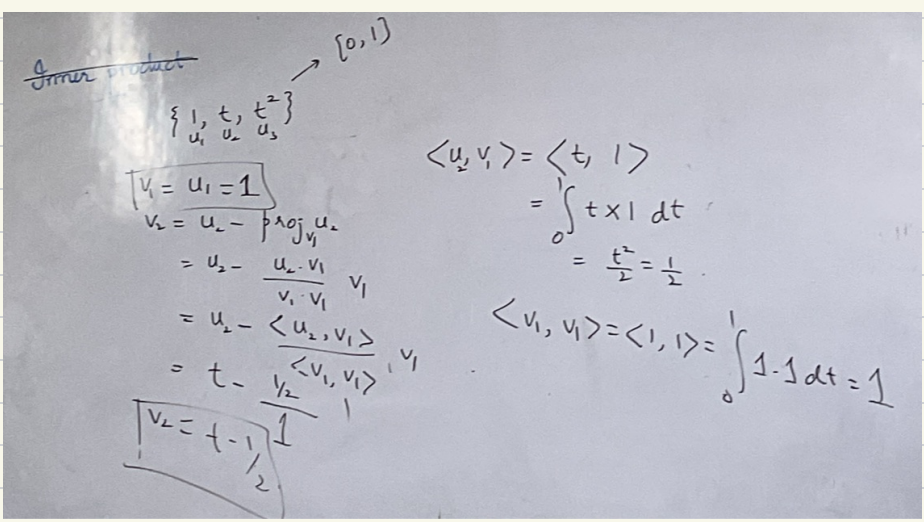
$\|\text{norm}\| = \sqrt{\text{trace}(A^T A)}$

↳ gram matrix

→ $\{1, t, t^2\}$ ⟶ $[0,1]$

check whether its orthogonal or not.

→ Gram-Schmidt conversion:-



$\{u, v, w\}$ → orthogonal set

$x = c_1u + c_2v + c_3w$ → if it is an orthogonal set

$x = \frac{x \cdot u}{u \cdot w}u + \frac{x \cdot v}{v \cdot v}v + \frac{x \cdot w}{w \cdot w}w$ → projection.

## Orthogonal Diagnolization:

→ $A = PDP^{-1}$

→ diagonalization of Real symmetric matrices

→ eigen values are always real

→ $A = PDP^{-1}$

↓

if they are orthogonal

for a orthogonal matrix $Q^{-1} = Q^T$

→ $A = QDQ^T$

→ eigen vectors of distinct eigen values are orthogonal

if repeated eigen value

if they are not orthogonal ↳ the use Gram Schmidth to make it orthogonal

orthonormal

$$= \begin{pmatrix} | & | & | \\ v_1 & v_2 & v_3 \\ | & | & | \end{pmatrix} \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{pmatrix} \begin{pmatrix} — v_1^T — \\ — v_2^T — \\ — v_3^T — \end{pmatrix}$$

$\lambda_1 v_1 v_1^T + \lambda_2 v_2 v_2^T + \lambda_3 v_3 v_3^T$

$\lambda_1 v_1 v_1^T$ → projection ↳ because norm is 1.

$\lambda_2 v_2 v_2^T$ → spectral decomposition (??)

$\lambda_3 v_3 v_3^T$

# Overview of HWs:

## Linear regression:-

$\hat{y} = Aw$

$(1 \quad x) \qquad (A^TA)^{-1}A^Ty$

for ridge $\rightarrow \left((A^TA + \alpha I)^{-1}A^T\right)y$

* weighted linear regression ..... → weights assign

* weighted

## Quadratic Form:-

$Q: \mathbb{R}^n \rightarrow \mathbb{R}$

$Q(x) = x^TAx$

$x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$ $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ → $x_1^2$, $x_1x_2$ (added), $x_2^2$

$= \begin{pmatrix} x_1 & x_2 \end{pmatrix}\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$

$= \begin{pmatrix} ax_1 + cx_2 & bx_1 + dx_2 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$

$= ax_1^2 + cx_1x_2 + bx_1x_2 + dx_2^2$

$x^TAx = ax_1^2 + (b+c)x_1x_2 + dx_2^2$

example:-

$Q = x_1^2 + x_2^2 + 7x_3^2 + 4x_1x_2 + 9x_1x_3 + 14x_1x_3$

$A = \begin{pmatrix} 1 & 2 & 9/2 \\ 2 & 1 & 7 \\ 9/2 & 7 & 7 \end{pmatrix}$ → $x_1$, $x_2$, $x_3$ (↓ $x_1$, $x_2$, $x_3$)

$Q(x_1, x_2) = \lambda_1(y_1)^2 + \lambda_2(y_2)^2$

$$A = \begin{pmatrix} 1 & 2 \\ 7 & 8 \end{pmatrix}$$

↳ $x_1^2 + 8x_2^2 + 9x_1x_2$

↳ can be written as:-

$x^2 + 9xy + 8y^2$

## Definite matrix

| | | | | |
|---|---|---|---|---|
| * | $x^TAx > 0, \quad \forall x \neq 0$ | A | positive definite | $\lambda_n > 0$ |
| * | $x^TAx \geq 0 \quad \forall x$ | A | semipositive definite | $\lambda_n \geq 0$ |
| * | $x^TAx < 0$ | | negative definite | $\lambda_n < 0$ |
| * | $x^TAx \gtrless 0$ | | indefinite | $\lambda_n \gtrless$ |

→ A should always be symmetric
→ we can make it symmetric by defining a matrix through quadratic form

**Sylvester's Criterion:** (to check definite matrix)

for a matrix A to be positive definite, all leading principle minors must be positive.

for a matrix A to be negative definite, the leading principle minor must alternate in sign, starting with negative values.

other than this, always indefinite.

$$A = \begin{bmatrix} 3 & 1 & 1 \\ 1 & 2 & 2 \\ 0 & 0 & 4 \end{bmatrix} \qquad [3] \quad \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix} \qquad \begin{bmatrix} 3 & 1 & 1 \\ 1 & 2 & 2 \\ 0 & 0 & 4 \end{bmatrix}$$

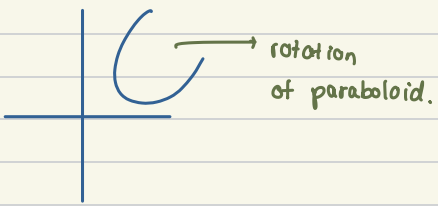
shows rotation.

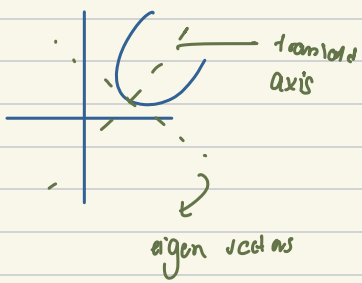
$ax^2 + by^2 + dx + ey + g + cxy = 0$

parabola, ellipse, circle, hyperbole, degenerate (parallel lines)

$ax^2 + by^2 + cz^2 + dx + ey + fz + hxy + iyz + jxz = 0$

paraboloid, ellipsoid, circloid, cone, huperboloid (1 sheet) (2 sheet), degenerate (11 planes)




geometric intrepetation of positive definite matrice.

→ $0 = 0$ → (minimum)

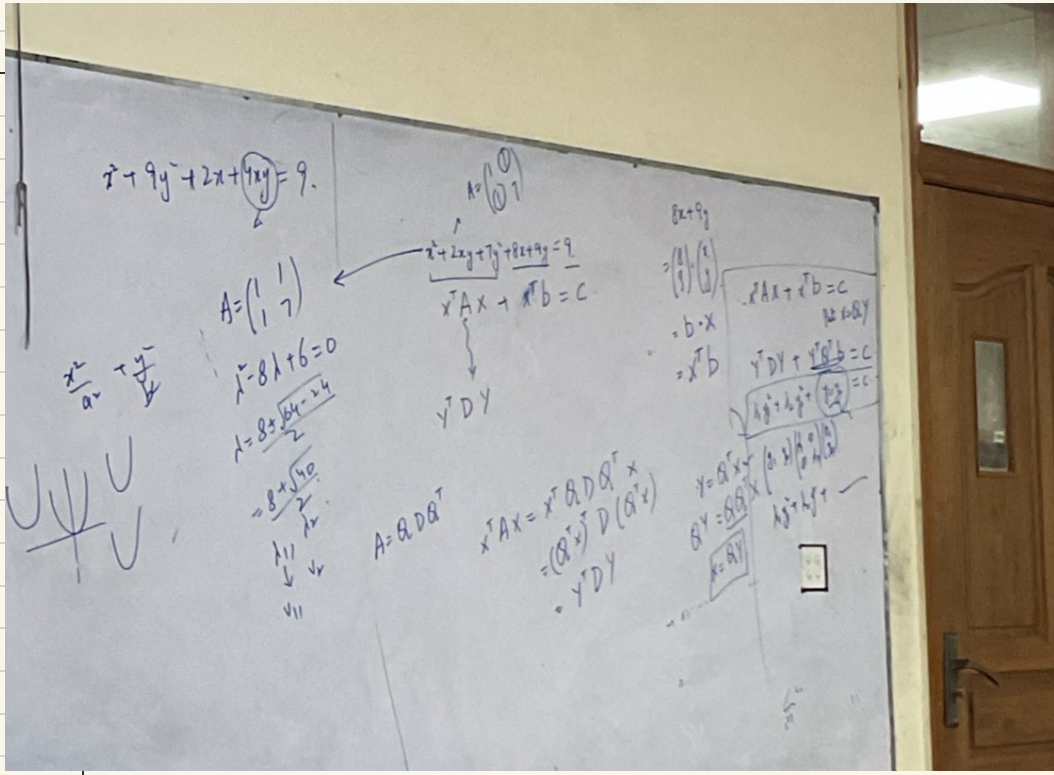
→ always postive

//

negative definite (maxima)

//

indefinite (saddle point)




hi muneeba ☺

how are you im fine thank you and you okay byebye

armish is so nice and awesome and she is the bestest person ever wow pen change cute color this is wow anyways cute clothes i said but you did not say thank you you are cancelled bye :v

Consider the following matrices

$$A = \begin{bmatrix} 6 & 2 & 1 \\ 2 & 6 & 2 \\ 1 & 2 & 6 \end{bmatrix}.$$

$$B = \begin{bmatrix} 4 & 1 & 2 \\ 1 & 4 & 1 \\ 2 & 1 & 4 \end{bmatrix}$$

- Find the eigenvalues of the matrices $A$ and $B$
- Find the eigenvectors corresponding to each eigenvalue.
- Check if the matrices are diagonalizable. ⟶ symmetric are always diagonalizable
- Perform the orthogonal diagonalization of $A$ and $B$.
- Write $A$ and $B$ as sum of Rank 1 projection matrices.

↳ spectral decomposition

eigen values — An $A_{n\times n}$ matrix is reduced as sum of Rank 1

$A_1 = \lambda_1 u_1 u_1^T + \lambda_2 u_2 u_2^T + \lambda_3 u_3 u_3^T$

matrix: A

$A - I\lambda = 0$

$$= \begin{bmatrix} 6-\lambda & 2 & 1 \\ 2 & 6-\lambda & 2 \\ 1 & 2 & 6-\lambda \end{bmatrix}$$

## Rayleigh Quotient (Contraint Optimization):

$$\frac{x^T A x}{x^T x}, \quad \|x\| \neq 0 \text{ or } x \neq 0$$

$$\max_{x \neq 0} \frac{x^T A x}{x^T x} = \lambda_{max}$$

$$\min_{x \neq 0} \frac{x^T A x}{x^T x} = \lambda_{min}$$

## SVD (Singular Value Decomposition:)

(latin symentic analysis, NLP)

$A_{m\times n} = U \Sigma V^T$ — $\Sigma$ → diagonal; $U$ → orthogonal

eigen decomposition (diagonalization) $\{A_{n\times n}\}$

spectral decomposition $\{A \rightarrow$ symmetric$\}$

$A^TA \rightarrow \sqrt{\lambda} \rightarrow \delta \rightarrow$ singular value

eigen vector of $A^TA \rightarrow V \rightarrow$ (mandatory)

eigen vector of $AA^T \rightarrow U \rightarrow$ (optional method) (Not recommended)

↳ $u_i = \frac{Av_i}{\delta i}$

PCA → (dimension reduction)

Curse of dimentionality

↳ Data Matrix

$x_1 = \begin{pmatrix} \\ \end{pmatrix}$, $x_2 = \begin{pmatrix} \\ \end{pmatrix}$ Data matrix $\begin{pmatrix} x_1^T \\ x_2^T \\ x_3^T \end{pmatrix}$

$x_3 = \begin{pmatrix} \\ \end{pmatrix}$, $x_4 = \begin{pmatrix} \\ \end{pmatrix}$ $D = \begin{pmatrix} F1 & F2 & F3 \\ \downarrow & \downarrow & \downarrow \end{pmatrix}$

$\vec{u} = \begin{pmatrix} \\ \end{pmatrix}$ ← data to be project on a line u, so that data's variance should be observed.

↳ (minimum info loss)

$\frac{u^T (D^T D) u}{u^T u}$ → Rayleigh Quotient

→ A

↳ calculate maxima

## Square Root of a matrix:-

as $\sqrt{9} = 3 \Rightarrow 3^2 = 9$

so in matrix we can write as

$\sqrt{B^2} = A$

using idea of Semi-Positive → $x^T A x \geq 0 \quad \forall x \in \mathbb{R}$

$\lambda_i \geq 0$

finding square root of a symmetric semi positive definite matrix.

$A = Q D Q^T \qquad D^{1/2} \cdot D^{1/2} = D$

$B = Q D^{1/2} Q^T \longrightarrow$ Square root of A

$B^2 = B \cdot B$

$= (Q D^{1/2} Q^T)(Q D^{1/2} Q^T)$

$= Q D^{1/2} Q^T Q D^{1/2} Q^T$

$= Q D^{1/2} D^{1/2} Q^T$

$B^2 = Q D Q^T$

$B^2 = A$

example:

$$A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$$

① $A^T = A$ (symmetric)

② $[1] \geq 0, \quad \left|\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}\right| \geq 0$ (leading principle minors) SPD

OR

$\lambda^2 - 5\lambda = 0$

$\lambda(\lambda - 5) = 0$

$\lambda = 0 \quad \lambda = 5 \qquad \lambda_i \geq 0 \quad \forall i = 1, 2$

$\begin{pmatrix} -2 \\ 1 \end{pmatrix} \quad \begin{pmatrix} 1 \\ 2 \end{pmatrix} \longrightarrow \begin{pmatrix} -2/\sqrt{5} \\ 1/\sqrt{5} \end{pmatrix} \cdot \begin{pmatrix} 1/\sqrt{5} \\ 2/\sqrt{5} \end{pmatrix}$

↓ orthonormal

$B = QD^{1/2}Q^T$

$$Q \begin{pmatrix} -2/\sqrt{5} & 1/\sqrt{5} \\ 1/\sqrt{5} & 2/\sqrt{5} \end{pmatrix}$$

$$D \begin{pmatrix} 0 & 0 \\ 0 & \sqrt{5} \end{pmatrix}$$

Rayleigh Quotient

Why do we need to study it?

* Principle Component Analysis:

$$\frac{x^T \Sigma x}{x^T x} \longrightarrow \max$$

* Linear Discriminant Analysis

$$\frac{x^T S_b x}{x^T S_w x} \longrightarrow \max$$

function:

$$f: \mathbb{R}^n - \{0\} \longrightarrow \mathbb{R}$$

$$f(x) = \frac{x^T A x}{x^T x}, \quad x \neq 0, \quad A \text{ is symmetric}$$

↗ Rayleigh Quotient

## Properties of Rayleigh Quotient.

$$f(x) = \frac{x^T A x}{x^T x}, \quad f(2x) = \frac{2x^T A 2x}{2x^T 2x} = \frac{4 x^T A x}{4 x^T x} = \frac{x^T A x}{x^T x}, \quad f(kx) = \frac{k x^T A k x}{k x^T k x} = \frac{k^2 x^T A x}{k^2 x^T x} = \frac{x^T A x}{x^T x}$$

$\Rightarrow$ Hence $f(kx) = \frac{x^T A x}{x^T x}$

$$f(kx) = f(x) \qquad k \neq 0$$

$$\frac{x^T A x}{x^T x} = \frac{x^T A x}{x \cdot x} = \frac{x^T A x}{\|x\|^2} = \left(\frac{x^T}{\|x\|}\right) A \left(\frac{x}{\|x\|}\right)$$

### forms of Rayleigh Quotient:

$\frac{x^T A x}{x^T x}, \; x \neq 0 \Rightarrow x^T A x \quad \|x\| = 1$

### example:

$$A = \begin{pmatrix} 1 & 3 \\ 3 & 2 \end{pmatrix}$$

$$f(x) = \frac{x^T A x}{x^T x} = \frac{x_1^2 + 6 x_1 x_2 + 2 x_2^2}{x_1^2 + x_2^2}, \quad x \neq 0$$

finding max / min value of Rayleigh Quotient.

$$\max_{x \neq 0} \frac{x^T A x}{x^T x} = \lambda_{max}$$

$$\min \frac{x^T A x}{x^T x} = \lambda_{min}$$

$$\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$$

$$\frac{x^T A x}{x^T x} = \frac{x^2 + 4xy + 4y^2}{x^2 + y^2}$$

$\lambda^2 - 5\lambda = 0$

$\lambda = 0, \quad \lambda = 5$

$(A - \lambda I) x = 0$

$(A - 0I) x = 0$

$$\begin{pmatrix} 1 & 2 & 0 \\ 2 & 4 & 0 \end{pmatrix}$$

$x + 2y = 0$

$x = -2y$

$$\begin{pmatrix} x \\ y \end{pmatrix} = y \begin{pmatrix} -2 \\ 1 \end{pmatrix}$$

$(A - \lambda I) x = 0$

$(A - 5I) x = 0$

$$\begin{pmatrix} -4 & 2 & 0 \\ 2 & -1 & 0 \end{pmatrix}$$

$-4x + 2y = 0$

$x = 2y$

$$\begin{pmatrix} x \\ y \end{pmatrix} = x \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

dividing by $\sqrt{5}$ or

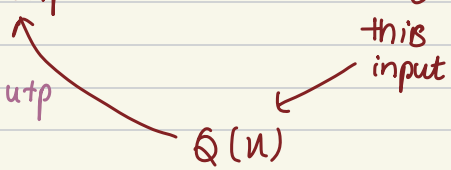
to orthonormalize them

Max value of $Q(x)$ is 5 which is achieved at $\frac{1}{\sqrt{5}}\begin{pmatrix}1\\2\end{pmatrix}$

Min value of $Q(x)$ is 0 which is achieved at $\frac{1}{\sqrt{5}}\begin{pmatrix}-2\\1\end{pmatrix}$

this output

this input




Proving theorem:-

**Theorem:** If a matrix is symmetric

$$\max_{x \neq 0} \frac{x^T A x}{x^T x} = \lambda_{max}$$

**Proof:**

$$A = QDQ^T$$

$$x^T A x = x^T (QDQ^T) x$$

$$= (x^T Q) D (Q^T x)$$

$$= (Q^T x)^T D (Q^T x)$$

$$= y^T D y$$

$$\|y\|^2 = y \cdot y = y^T y = x^T Q Q^T x = x^T x = 1$$

$$x^T A x = y^T D y$$

Quadratic form:

$$y^T D y = \sum_{i=1}^{n} \lambda_i y_i^2$$

## SVD (Singular Value Decomposition):

* $A = U \Sigma V^T$

applications:
- → used in dimension reduction
- → // noise // //
- → data compression
- → ill conditioned systems → (??)
- → latent semantic analysis
- → face recognition.

* four fundamental Spaces
  - Col(A)
  - Row(A)
  - Null(A)
  - Null($A^T$)

Recalling
- $A_{n \times n} = PDP^{-1}$ → square
  - → diagonalization
  - → eigen decomposition
- $A_{n \times n} = QDQ^T$ → symmetric
  - → Spectral decomposition
  - → orthogonal decomposition.

SVD:-

$$A_{m \times n} = U_{m \times m} \; \Sigma_{m \times n} \; V^T_{n \times n}$$

- U: rotation; orthonormal column; length & angle preserving
- Σ: scaling; singular values on diagonal ($\sqrt{\lambda}$)
- $V^T$: rotation; orthonormal column

$$A^TA = (U\Sigma V)^T(U\Sigma V)$$

$$= V\,\Sigma^T U^T U\,\Sigma V^T$$

$U^TU$ → orthogonal

$$A^TA = V\,\Sigma^T\Sigma V^T$$

$\Sigma^T\Sigma$ → $\sqrt{\text{eigen value of } A^TA}$

$V$ → set of orthonormal eigen vectors of $A^TA$

$$(A^TA)^T = A^T(A^T)^T$$
$$= A^TA$$

as $(A^TA)^T = A^TA$ so it is symmetric matrix.

$$A = Q\,\Lambda\,Q^T$$

$$AA^T = (U\Sigma V^T)(U\Sigma V^T)^T$$
$$= U\Sigma V^T V\Sigma^T U^T$$
$$= U\Sigma\Sigma^T U^T$$

$A = U\Sigma V^T$ → singular vallue decomposition

- $V$ → eigen vectors of $A^TA$
- $U$ → eigenvector of $AA^T$

$$A = U\Sigma V^T$$
$$AV = U\Sigma V^TV$$
$$AV = U\Sigma$$ → used to calculate $U$

opening up:-

$$Av_i = \sigma_i u_i$$
$$u_i = \frac{1}{\sigma_i} Av_i$$

$\sigma_i$ → singular value $= \sqrt{\lambda}$

# Singular Value Decomposition

example 1:-

$$A = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{bmatrix}$$

→ find SVD of A

→ $A = U \Sigma V^T$

→ We can calculate $AA^T$ and write the eigen values of $A^TA$.

$$AA^T = \begin{pmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{pmatrix} \begin{pmatrix} 3 & 2 \\ 2 & 3 \\ 2 & -2 \end{pmatrix}$$

$$= \begin{pmatrix} 17 & 8 \\ 8 & 17 \end{pmatrix}$$

$$\lambda^2 - 34\lambda + 225 = 0$$

$$(\lambda - 25)(\lambda - 9) = 0$$

$AA^T \longrightarrow \lambda_1 = 25, 9$

$A^TA \longrightarrow \lambda_1 = 25, 9, 0$

$\sigma_1 = \sqrt{25} = 5$ ⟶ it contains more info about the data.

$\sigma_2 = \sqrt{9} = 3$

* $A^TA$

$= x^T A^T A x$

$= (Ax)^T Ax$

$= Ax \cdot Ax$

$= \|Ax\|^2 \geq 0$

$x^T (A^TA) x \geq 0$

$A^TA$ is Semi Positive Definite

$\lambda_i \geq 0$

* all non-zero eigen values of $A^TA$ and $AA^T$ are same

↳ $A_{2\times3}$

$(A^TA)_{3\times3} = \lambda_1, \lambda_2, \lambda_3$ (always zero)

$(AA^T)_{2\times2} = \lambda_1, \lambda_2$

↳ why eigen value is zero?

⟹ because A is an invertible matrix.

calculating $A^TA$

$A^TA \longrightarrow 25 \quad (A^TA - 25I)x = 0$

$\longrightarrow 9 \quad (A^TA - 9I)x = 0$

$\longrightarrow 0 \quad (A^TA - 0I)x = 0$

25 9 0

$$\frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\\0\end{pmatrix}, \frac{1}{\sqrt{18}}\begin{pmatrix}1\\-1\\4\end{pmatrix}, \frac{2}{3}\begin{pmatrix}1\\-1\\-1/2\end{pmatrix}$$

set of orthonormal eigen vectors.

$$V = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{18} & 2/3 \\ 1/\sqrt{2} & -1/\sqrt{18} & -2/3 \\ 0 & 4/\sqrt{18} & -1/3 \end{bmatrix}$$

set of singular values on the diagonal

$$\Sigma = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 3 & 0 \end{bmatrix}$$

$$A_{2\times3} = U_{2\times2}\ \Sigma_{2\times3}\ V^T_{3\times3}$$

$$u_1 = \frac{1}{\sigma_1} A v_1 = \frac{1}{5}\begin{pmatrix}3 & 2 & 2\\ 2 & 3 & -2\end{pmatrix}\begin{pmatrix}1/\sqrt{2}\\1/\sqrt{2}\\0\end{pmatrix} = \begin{pmatrix}1/\sqrt{2}\\1/\sqrt{2}\end{pmatrix}$$

$$u_2 = \frac{1}{\sigma_2} A v_2 = \frac{1}{3}\begin{pmatrix}3 & 2 & 2\\ 2 & 3 & -2\end{pmatrix}\begin{pmatrix}1/\sqrt{18}\\-1/\sqrt{18}\\4/\sqrt{18}\end{pmatrix} = \begin{pmatrix}1/\sqrt{2}\\-1/\sqrt{2}\end{pmatrix}$$

$$U = \begin{pmatrix}1/\sqrt{2} & 1/\sqrt{2}\\ 1/\sqrt{2} & -1/\sqrt{2}\end{pmatrix}$$

## Example 2:

Find SVD: $A = \begin{pmatrix} 1 & 0 & 1 \\ -1 & 1 & 0 \end{pmatrix}_{2\times 3}$

$A_{2\times 2} = U_{2\times 2}\, \Sigma_{2\times 3}\, V^T_{3\times 3}$

$$AA^T = \begin{pmatrix} 1 & 0 & 1 \\ -1 & 1 & 0 \end{pmatrix}\begin{pmatrix} 1 & -1 \\ 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$$

finding eigenvalues.

$\lambda^2 - 4\lambda + 3$

$(\lambda - 3)(\lambda - 1)$

$AA^T \longrightarrow \lambda \to 3, 1$

$A^TA \longrightarrow \lambda \to 3, 1, 0$

$\delta_1 = \sqrt{3}$

$\delta_2 = 1$

$$\Sigma = \begin{pmatrix} \sqrt{3} & 0 & 0 \\ 0 & \sqrt{1} & 0 \end{pmatrix}$$

finding eigen vectors & making orthonormal.

$3 \to \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} \frac{1}{\sqrt{6}}$

$1 \to \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \frac{1}{\sqrt{2}}$

$0 \to \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix} \frac{1}{\sqrt{3}}$

$$V = \begin{pmatrix} 2/\sqrt{6} & 0 & -1/\sqrt{3} \\ -1/\sqrt{6} & 1/\sqrt{2} & -1/\sqrt{3} \\ 1/\sqrt{6} & 1/\sqrt{2} & 1/\sqrt{3} \end{pmatrix}$$

$$u_1 = \frac{Av_1}{\delta_1} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$u_2 = \frac{Av_2}{\delta_2} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$U = \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{pmatrix}$$

Example #3:-

Find SVD

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{pmatrix}_{3\times 2}$$

$$A_{3\times 2} = U_{3\times 3} \Sigma_{3\times 2} V^T_{2\times 2}$$

$$A^TA = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

$$\lambda^2 - 4\lambda + 3 = 0$$

$$\lambda = 3, 1$$

$$\sigma_1 = \sqrt{3}, \quad \sigma_2 = 1$$

$$\Sigma = \begin{pmatrix} \sqrt{3} & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$$

$$\lambda = 3 \quad = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\lambda = 1 \quad = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$V = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix}$$

$$u_1 = \frac{Av_1}{\sigma_1}, \quad u_2 = \frac{Av_2}{\sigma_2}, \quad u_3 = \frac{Av_3}{\sigma_3}$$

$$\begin{pmatrix} \sqrt{3}/3 \\ \sqrt{6}/6 \\ \sqrt{6}/6 \end{pmatrix} \qquad \begin{pmatrix} 0 \\ -\sqrt{2}/2 \\ \sqrt{2}/2 \end{pmatrix} \qquad \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

↳ $u_3 \cdot u_1 = 0 \qquad u_3 \cdot u_2 = 0$

$u_3 \cdot u_1 = 0$

$\frac{\sqrt{6}}{3}x + \frac{\sqrt{6}}{6}y + \frac{\sqrt{6}}{6}z = 0$ $\qquad$ $-\frac{\sqrt{2}}{2}y + \frac{\sqrt{2}}{2}z = 0$

$\frac{\sqrt{6}}{3}x + \frac{\sqrt{6}}{3}z = 0$ $\qquad$ $-\frac{\sqrt{2}}{2}y = -\frac{\sqrt{2}}{2}z$

$\frac{\sqrt{6}}{3}x = -\frac{\sqrt{6}}{3}z$ $\qquad$ $y = z$

$x = -z$

$z\begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix} \longrightarrow \begin{pmatrix} -1/\sqrt{3} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{pmatrix} \rightarrow u_3$

## SVD as sum of rank 1 matrix:-

$A = U \Sigma V^T$

generalized spectral decomposition.

$A = \sigma_1 u_1 v_1^T + \sigma_2 u_2 v_2^T + \cdots + \sigma_r u_r v_r^T$

↳ rank 1 matrice

↙

$u_i v_i^T \longrightarrow$ dot product

↳ any matrix that is

rectangular / square / symmetric / non-symmetric

$\sigma_r$ → rank of given matrix.

LA → (Quiz)

Data → (Quiz)

# SVD and basis of 4 fundamental Subspaces

$A = U \Sigma V^T$
$m \times n \quad m \times m \quad m \times n \quad n \times n$

+ orthogonal
+ square matrix

Subspaces.
- Row Space (A)
- Col (A)
- Null (A) $(Ax = 0)$
- Left Null (A) $(A^T x = 0)$

**fact 1:-**

$A = \begin{pmatrix} 1 & 1 \\ 0 & -1 \\ 1 & 1 \end{pmatrix}_{3 \times 2}$

Row Space → Span of Rows → $\mathbb{R}_2$

Column Space → Span of Cols → $\mathbb{R}_3$

Null Space → $Ax = 0$ ($3\times2$, $2\times1$) → $\mathbb{R}^2$

Left Null Space → $A^T x = 0$ ($2\times3$, $3\times1$) → $\mathbb{R}^3$

**fact 2:**

$A_{3\times2} = U_{3\times3} \; \Sigma_{3\times2} \; V^T_{2\times2}$

$U$: ColSpace, Left NullSpace

$V^T$: RowSpace, NullSpace

**fact 3:**

Rank (A) =
- no. of non-zero rows.
- Dim of Col(A)
- Dim of Row(A)
- no. of non-zero singular values
  - $\sqrt{\lambda_i} \rightarrow A^TA$ (or $AA^T$)

**fact 4:**

$A = \sigma_1 u_1 v_1^T + \sigma_2 u v_2^T + \dots + \sigma_r u_r v_r^T$ → rank of matrix.

## Collary :-

- $u_1, \ldots, u_r$ ⟶ orthonormal bases of Col (A)
- $u_{r+1}, \ldots, u_m$ ⟶ // // Left Null (A)
- $v_1, \ldots, v_r$ ⟶ // // Row (A)
- $v_{r+1}, \ldots, v_n$ ⟶ // // Null (A)

**Example:**

$$A = \begin{pmatrix} 3 & 0 \\ 4 & 5 \end{pmatrix}$$

$$A^TA = \begin{pmatrix} 25 & 20 \\ 20 & 25 \end{pmatrix}$$

$\lambda_1 = 45$ , $\lambda_2 = 5$

$\sigma_1 = \sqrt{45}$ $\quad$ $\sigma_2 = \sqrt{5}$ $\quad$ rank = 2

$v_1 = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ , $v_2 = \frac{1}{\sqrt{2}}\begin{pmatrix} -1 \\ 1 \end{pmatrix}$

$v_i$'s are called right singular vectors.

$u_1 = \frac{Av_1}{\sigma_1}$ , $u_2 = \frac{Av_2}{\sigma_2}$ $\quad$ $u_i$'s are called left singular vectors.

Col (A) ⟶ rank = 2

$$U = \frac{1}{\sqrt{10}}\begin{pmatrix} 1 & 3 \\ 3 & 1 \end{pmatrix} \qquad \Sigma = \begin{pmatrix} \sqrt{45} & 0 \\ 0 & \sqrt{5} \end{pmatrix}$$

Left Null (A) = 0

Row (A) ⟹ rank = 2

$$V = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

Nullspace (A) = 0

## SVD and matrix norms:

→ Frobenius Norm: (magnitude)

$$\|A\|_F = \sqrt{\sum_{i=1}^{r} \sigma_i^2}$$

→ Spectral Norm: (maximum stretch)

$$\|A\|_F = \sigma_1 \text{ (largest)}$$

→ Nuclear Norm: (total size of singular values)

$$\|A\|_N = \sum_{i=1}^{r} \sigma_i$$ → (for writing matrix into low rank matrices)

→ **Application of SVD:-**

→ Principal Component Analysis (PCA):

Purpose:

for projecting the data from higher dimension to lower dimension.

→ Dimension Reduction:

→ reduces dimesion, by finding variable that are uncorelated

→ they retain most of variation of data.

→ Feature Extraction:

→ finding most important feature → of maximum variance

→ Noise Reduction:

→ distorted patterns

→ ignore the noise of data.

→ Image Recognition & Objection Detection

→ NLP → Semantic Latent

→ Reinforcement Learning.

* Calculate mean

$$x = \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix}$$

$$\text{mean} = \mu = \frac{1+2+3+4}{4} = \frac{10}{4} = \frac{5}{2} = 2.5$$

* Data matrix

$$x_1 = \begin{pmatrix} a_1 \\ b_1 \\ c_1 \\ d_1 \end{pmatrix}, \quad x_2 = \begin{pmatrix} a_2 \\ b_2 \\ c_2 \\ d_2 \end{pmatrix}, \quad x_3 = \begin{pmatrix} a_3 \\ b_3 \\ c_3 \\ d_3 \end{pmatrix}$$

$$D = \begin{pmatrix} \text{——} x_1^T \text{——} \\ \text{——} x_2^T \text{——} \\ \text{——} x_4^T \text{——} \end{pmatrix}$$

$$D = \begin{pmatrix} a_1 & b_1 & c_1 & d_1 \\ a_2 & b_2 & c_2 & d_2 \\ a_3 & b_3 & c_3 & d_3 \end{pmatrix}$$

* Centered Data Matrix

$D \rightarrow$ data matrix

$$\bar{D} = D - 1 \cdot \mu^T \quad \left(\text{Centered Data matrix}\right)$$

* Projection:

$u$ ⟶ vector along which we want to project our data

→ The projection of the centered point.

$\bar{x_i} \in \bar{D}$ ⟶ $u$ ⟶ if unit vector

$$\text{Proj}_u \bar{x_i} = \left(\frac{\bar{x_i} \cdot u}{u \cdot u}\right) u = \left(\frac{u^T \cdot \bar{x_i}}{u^T u}\right) u$$

$$\hookrightarrow (u^T \bar{x_i}) u$$

$$= a_i \times u$$

$$a_i = u^T \bar{x_i}$$

$i = 1, 2, 3 \ldots n$ (data points)

* mean of the projected points:

$$a = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix}$$

$$\mu_a = \frac{a_1 + a_2 + \ldots a_n}{n}$$

$$= \frac{1}{n} \sum_{i=1}^{n} a_i$$

$$= \frac{1}{n} \sum_{i=1}^{n} u^T \bar{x_i}$$

$$= u^T \left(\frac{\sum_{i=1}^{n} \bar{x_i}}{n}\right)$$

↳ equals to mean

$$= u^T \bar{\mu}$$

$$= 0$$

*

Variance:

$$\sigma_u^2 = \frac{1}{n}\sum_{i=1}^{n}(a_i - \mu_a)^2 = \frac{1}{n}\sum_{i=1}^{n} a_i^2$$

$$= \frac{1}{n}\sum_{i=1}^{n}(u^T \bar{x_i})^2$$

$$= \frac{1}{n}\sum_{i=1}^{n} u^T (\bar{x_i}\bar{x_i}^T) u$$

$$= u^T \left(\frac{\sum_{i=1}^{n} \bar{x_i}\bar{x_i}^T}{n}\right) u$$

$\Sigma \rightarrow$ covariance

$$\boxed{\sigma_u^2 = u^T \Sigma u}$$

we want to maximize the variance. ⟶ at maximum eigen value.

finding maxima $u^T \Sigma u$, $\|u\| = 1$

↳ Rayleigh Quotient.

↳ $x^T A x = \|x\| = 1$

↳ max $\lambda_i \rightarrow v_i$

$$\Sigma = \frac{1}{n}\sum_{i=1}^{n} \bar{x_i}\bar{x_i}^T$$

$$C = \begin{pmatrix} \text{Var}(x) & \text{Cov}(x,y) \\ \text{Cov}(x,y) & \text{Var}(y) \end{pmatrix}_{2\times 2}$$

Gram matrix:

$$A^T A \rightarrow \frac{1}{(n-1)} \rightarrow \text{vessel correction} \quad (\bar{D}^T \bar{D})$$

$$\boxed{C = \frac{1}{n-1} (\bar{D}^T \bar{D})} \rightarrow \lambda_{max} \hookrightarrow v_i$$

the direction of vector u should be the direction of the eigen vector corresponding to the maximum eigen value.

Example:-

Problem:

Suppose we have the following 2D-data set

| Sample | $x_1$ | $x_2$ |
|---|---|---|
| 1 | 2 | 4 |
| 2 | 0 | 0 |
| 3 | 1 | 1 |
| 4 | 3 | 3 |

→ Step 1:

write data matrix:-

$$D = \begin{pmatrix} — x_1^T — \\ — x_2^T — \\ — x_3^T — \\ — x_4^T — \end{pmatrix}$$

$$D = \begin{pmatrix} 2 & 4 \\ 0 & 0 \\ 1 & 1 \\ 3 & 3 \end{pmatrix}$$

4 sample points

4 × 2 → 2 features

→ Step 2: find mean:

$$\mu_1 = \frac{2+0+1+3}{4} = \frac{6}{4} = 1.5$$

$$\mu_2 = \frac{4+0+1+3}{4} = 2$$

→ Step 3: centralize the data matrix

$$\bar{D} = \begin{pmatrix} 2-1.5 & 4-2 \\ 0-1.5 & 0-2 \\ 1-1.5 & 1-2 \\ 3-1.5 & 3-2 \end{pmatrix} = \begin{pmatrix} 0.5 & 2 \\ -1.5 & -2 \\ 0.5 & -1 \\ 1.5 & 1 \end{pmatrix}$$

→ Step 4: find Covariance matrix, which is actually gram matrix

$$\Sigma = C = \frac{1}{4-1} \bar{D}^T \bar{D} = \begin{pmatrix} 1.67 & 1.83 \\ 1.83 & 3.33 \end{pmatrix}$$

→ Step 5: find eigen values & eigen vectors.

$\lambda_1 = 4.97 \longrightarrow \begin{pmatrix} 0.66 \\ 0.75 \end{pmatrix}$

$\lambda_2 = 0.03$

$\downarrow$

$\begin{pmatrix} -0.75 \\ 0.66 \end{pmatrix}$

→ Step 6:

Total variance = trace of Covariance matrix

trace = $\sum \lambda_i$

→ Step 7:

finding principal components

$\lambda_{max} = \frac{4.97}{4.97 + 0.03} = \frac{4.97}{5} \times 100$

$= 99.4\%$

$\begin{pmatrix} 0.66 \\ 0.75 \end{pmatrix}$ ← if we project the data on this principle component 99.4% variance is preserved.

(latent semantic analysis:)

→ Step 8:

project the data:

Projection = $X_{centered}$ $V_1$

$$= \begin{pmatrix} 0.5 & 2 \\ -1.5 & -2 \\ -0.5 & -1 \\ 1.5 & 1 \end{pmatrix} \begin{pmatrix} 0.66 \\ 0.75 \end{pmatrix}$$

$$= \begin{pmatrix} 1.75 \\ -2.66 \\ -1.08 \\ 2.00 \end{pmatrix}$$

Recalling lecture:

* Remembering Projection:-

projection

- proj line: $\frac{u u^T}{u^T u}$
- proj plane: Bases $\{u, v\}$

$A^{-1}_{left} A = I$

$\left((A^T A)^{-1} A^T\right)_{left} A = I$

projection:

$P = A (A^T A)^{-1} A^T$

→ ColumnSpace

$A A^{-1}_{right} = I$

$A \left(A^T (A A^T)^{-1}\right)_{right} = I$

projection:

$P = A^T (A A^T)^{-1} A$

→ RowSpace

→

→ Orthogonal Diagonalization

→ Rayleigh Quotient

## Affine Combinations:

⇒ $\{x_1, x_2, \ldots, x_k\} \in \mathbb{R}^n$

⇒ $\sum_{i=1}^{k} c_i x_i$ ⇒ sum of linear combination.

↳ ⇒ sum of all $c = 1$

$$\sum_{i=1}^{k} c_i = 1$$

## Convex Combinations:

↳ exactly affine combination

↳ $c_i \geq 0$

→ **Affine Independence:-**

→ If the vectors are affinely independent then no vector can be written in the affine combination of the other vector.

→ **Affine null:-**

Set of all affine combination is called affine

**Theorem:-**

A point $y$ in $\mathbb{R}^n$ is an affine combination, of $v_1 \longrightarrow v_p$ in $R^n$ if and only if $y - v_1$ is linear combination of $v_2 - v_1$, $v_3 - v_i$ ... $v_3 - v_i$

A translate of a set in $\mathbb{R}^n$ by a vector $p$ is

$$S+p = \{ s+p : s \in S \}$$

→ A flat is a translation of subspace.

→ The dimension of the flat is the dimension of subspace.